

## Unit 9

# ALGORITHM DESIGN AND PROBLEM-SOLVING

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73 items · 382 marks

### **Syllabus sections in this unit**

- 9.1 Computational Thinking Skills
- 9.2 Algorithms

## Unit 9: Algorithm Design and Problem-solving

### 1. M/J 26/P21/Q2/a(ii,iii)

2 A washing machine contains:

- two trays for the washing powder
- a user interface display
- five sensors.

The washing machine control program includes the following variables:

| variable  | description                                            |
|-----------|--------------------------------------------------------|
| MainsOn   | TRUE when the mains power is on                        |
| WaterOn   | TRUE when the water supply to the machine is turned on |
| DisplayOn | TRUE when the user interface display is active         |
| Tray1     | TRUE when it detects washing powder in tray 1          |
| Tray2     | TRUE when it detects washing powder in tray 2          |

- (ii) The display is only active when both the mains power is on and the water supply is turned on.

Complete the pseudocode statement.

DisplayOn ←

[1]

- (iii) The washing machine control program will start when a new variable `WashStart` is assigned the value `TRUE`

`WashStart` is assigned the value `TRUE` when the mains power is switched on and there is washing powder in tray 1 or tray 2 or both trays.

Write the pseudocode statement for the given description.

[3]

**2. M/J 26/P21/Q4**

**4** An algorithm is expressed as:

- A user inputs a sequence of integer values.
- The process will calculate the number of positive integers.
- The user will input value 0 to indicate there are no more numbers to be input.
- Output the number of positive integers after the terminating 0 is input.

Write a structured English description for this algorithm as a sequence of steps.

Do **not** include pseudocode statements in your description.

Your answer might **not** require all **six** steps shown.

step 1

step 2

step 3

step 4

step 5

step 6

[4]

**3. M/J 26/P22/Q2/a,b,c**

**2** A new computer system is being developed to help manage a tool hire business. A customer can specify if they want to hire a tool for either half a day or a full day.

An abstract model for the new system is required.

**(a)** Explain the purpose of abstraction.

[1]

**(b)** State **four** items of data that must be stored each time a tool is hired.

item 1

item 2

item 3

item 4

[2]

**(c)** The computer system will have to process the tool hire data. One process is to cancel a tool hire booking.

State **two** other processes that would be required.

process 1

process 2

[2]

**4. M/J 26/P22/Q3**

- 3 A computer game uses a text file named `WordList.txt` containing 10 000 words, all in lower case. Each word is unique and is stored on a separate line of the text file.

An efficient algorithm is needed to check that a word input by the user is stored in `WordList.txt`. If the word is found, the line number where the word is stored must be output. If the word is **not** found, the message "Not Found" must be output.

Complete the steps needed to describe the algorithm.

Do **not** include pseudocode statements in your answer.

The first **three** steps have been completed for you.

step 1: set line counter to 0

step 2: input a word

step 3: convert the word to lower case

[6]

**5. M/J 26/P23/Q2/b,c**

- 2 A control program is being developed for a robot floor vacuum cleaner. The settings for the robot can be changed and include:

- the strength of the suction (numerical values: 0, 1, 2, 3, 4 or 5)
- the total vacuum cleaning time (a whole number of minutes).

The robot has a number of sensors.

The control program includes the variables:

| variable  | description                                                                                               |
|-----------|-----------------------------------------------------------------------------------------------------------|
| MainsOn   | TRUE when the mains power is on                                                                           |
| DisplayOn | TRUE when the user display is shown                                                                       |
| Suction   | TRUE when the program detects that the user has selected a non-zero value for the strength of the suction |

- (b) Write a pseudocode statement to set the `DisplayOn` variable to TRUE when the mains power is on.

[1]

- (c) When the robot is ready to move, the program assigns the value TRUE to a variable, `Move`

The control program contains the pseudocode logic expression:

`Move ← MainsOn AND Suction`

Explain the pseudocode statement.

[2]

**6. M/J 26/P23/Q4**

**4** An algorithm is expressed as:

- A user inputs a sequence of integers.
- The user will input value  $-1$  to indicate there are no more integers to be input.
- The program will calculate the total number of integers greater than 120.
- Output the total number of integers greater than 120 after the  $-1$  terminator is input.

Write a structured English description for this algorithm as a sequence of steps.

Do **not** include pseudocode statements in your description.

Your answer might **not** require all **six** steps shown.

step 1

step 2

step 3

step 4

step 5

step 6

[4]

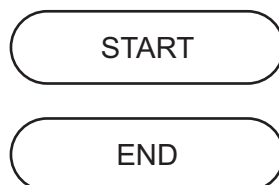
**7. O/N 25/P21/Q2**

**2** Data is a global 1D array containing 30 elements of type `STRING`

An algorithm will output:

- all non-blank elements (elements that do **not** contain an empty string)
- the final total of the number of elements output.

Complete the program flowchart to represent the algorithm:



[5]

**8. O/N 25/P22/Q1/a**

1 (a) The table contains pseudocode examples.

Each example may contain statements that relate to one or more of:

- selection
- iteration (repetition)
- subroutines (procedures or functions).

Complete the table by placing **one or more** ticks (✓) in each row.

| Pseudocode example                                                                         | Selection | Iteration | Subroutine |
|--------------------------------------------------------------------------------------------|-----------|-----------|------------|
| IF Status = FALSE THEN<br>FOR Count ← 1 TO 20<br>CALL Reset (Count)<br>NEXT Count<br>ENDIF |           |           |            |
| OTHERWISE : Status ← TRUE                                                                  |           |           |            |
| WHILE AllDone () = TRUE                                                                    |           |           |            |
| 30 : NextChar ← 'X'                                                                        |           |           |            |

[4]

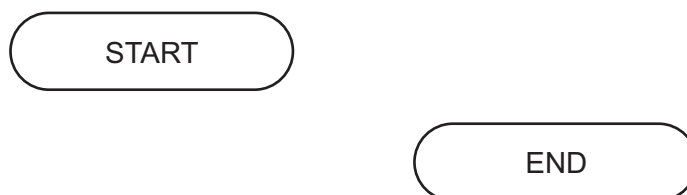
**9. O/N 25/P22/Q2**

2 Data is a global 1D array containing 20 elements of type REAL

An algorithm will:

- input a sequence of real values, one at a time
- assign each value to consecutive array elements, starting from index 1
- end when the value 99.9 is input, or all 20 elements have been assigned (the value 99.9 must **not** be stored in the array).

Complete the program flowchart to represent the algorithm:



[5]

**10. O/N 25/P22/Q3/a**

- 3 A text file `OldFile.txt` contains IDs, names and email addresses for members of a club. Three information items are stored for each member and each item is stored on a separate line.

The example shows the information for the first two members in the first six lines of the file:

| Line in file | Information item       | Example data                |
|--------------|------------------------|-----------------------------|
| 1            | member 1 ID            | "AB1234"                    |
| 2            | member 1 name          | "Freddie Jones"             |
| 3            | member 1 email address | "FreddieJ909@Cambridge.org" |
| 4            | member 2 ID            | "BC2345"                    |
| 5            | member 2 name          | "Sue Smith"                 |
| 6            | member 2 email address | "Sue1024@Cambridge.org"     |

The member ID string is always two letters followed by four digits.

The file design is to be changed so that information for each user is stored as a single line of the file, with the character '`\`' used as a separator between data items.

For example, the single line for member 1 will be:

"AB1234\Freddie Jones\FreddieJ909@Cambridge.org"

An algorithm will produce a new file `NewFile.txt` from the contents of `OldFile.txt`

Assume:

- `LineX`, `LineY` and `LineZ` are of type `STRING` and are used to store the three items of information for each member
- `NewString` is a temporary variable of type `STRING`
- `OldFile.txt` exists and contains valid data.

- (a) The algorithm to create the new file is expressed in steps.

Complete the following numbered steps:

- Step 1 :    open the file ..... in .....
- Step 2 :    open the file ..... in .....
- Step 3 :    read a line from ..... and store in
- Step 4 :    read a line from ..... and store in
- Step 5 :    read a line from ..... and store in
- Step 6 :    set `NewString` to
- Step 7 :    write `NewString` to .....
- Step 8 :    repeat from step ..... until .....
- Step 9 :    close both files.

**11. O/N 25/P22/Q4/c**

**4** A quiz has nine questions. There are:

- five easy questions each worth 3 points
- four hard questions each worth 5 points.

At the end of the quiz the points for each correctly answered question are added to give a total score.

A check is made to test that the total score is valid using a 1D array `CheckTotal` of type `BOOLEAN`. Each index value of the array represents a possible total score. The corresponding element value is `TRUE` if the index value represents a valid total score and `FALSE` otherwise.

The first nine rows of the array are:

| Index value | Element value | Comment                                                                   |
|-------------|---------------|---------------------------------------------------------------------------|
| 0           | TRUE          | valid total score (no correct answers)                                    |
| 1           | FALSE         | invalid total score                                                       |
| 2           | FALSE         | invalid total score                                                       |
| 3           | TRUE          | valid total score (one 3-point question correct)                          |
| 4           | FALSE         | invalid total score                                                       |
| 5           | TRUE          | valid total score (one 5-point question correct)                          |
| 6           | TRUE          | valid total score (two 3-point questions correct)                         |
| 7           | FALSE         | invalid total score                                                       |
| 8           | TRUE          | valid total score (one 3-point question and one 5-point question correct) |

For example, a total score of 6 points is valid; the value of the array element at index value 6 is `TRUE`

**(c)** The array elements have been assigned the required values.

A module `ValidateScore()` will take an integer value representing a total score as a parameter. It will return `TRUE` if the total score is valid, or `FALSE` if it is **not** valid.

Describe the algorithm for `ValidateScore()` using **four** steps.

Do **not** use pseudocode in your answer.

Step 1

Step 2

Step 3

Step 4

[4]

**12. O/N 25/P23/Q2/a,b**

- 2 (a)** A program contains a global 1D array of type `STRING` containing 65 elements.

An existing text file is used to store the data in the array. Only non-blank elements (those that do **not** contain an empty string) are written to the text file.

An algorithm will:

- write the first element of the array as a new line in the text file
- continue until all elements have been written, each to a new line of the text file.

Stepwise refinement is applied to the algorithm.

Describe up to **six** steps for this algorithm that could be used to produce pseudocode.

Do **not** use pseudocode statements in your answer.

Step 1

Step 2

Step 3

Step 4

Step 5

Step 6

[6]

- (b)** Iteration is one programming construct.

Identify **one** other programming construct that will be required when the algorithm from part **(a)** is converted into pseudocode and explain how it is used.

Construct

Use

[2]

**13. M/J 25/P21/Q1/a,b,c**

**1** A program is being developed to help the manager of a shop control the stock.

**(a)** An identifier table has been used during the design stage.

Complete the identifier table:

| Example value | Explanation                                  | Variable name | Data type |
|---------------|----------------------------------------------|---------------|-----------|
| "Fruit"       | a category of stock that is sold in the shop |               |           |
| 20/02/2025    | when an item was sold                        |               |           |
| 12.67         | the cost of an item                          |               |           |
| TRUE          | to indicate if an item is in stock           |               |           |

[4]

**(b)** A module `Sales()` is part of the stock control program.

The table contains pseudocode extracts from the module `Sales()`

Each extract may include all or part of:

- assignment
- selection
- iteration (repetition).

Complete the table by placing one or more ticks (✓) in each row:

| Pseudocode extract                                                            | Assignment | Selection | Iteration |
|-------------------------------------------------------------------------------|------------|-----------|-----------|
| <code>Result ← CalculateTotal()</code>                                        |            |           |           |
| <code>WHILE IsClosed</code>                                                   |            |           |           |
| <code>REPEAT<br/>  INPUT Value<br/>UNTIL Sales[4] &gt; Value</code>           |            |           |           |
| <code>IF Sales[Current] &lt;= 150 THEN<br/>  Discount ← TRUE<br/>ENDIF</code> |            |           |           |
| <code>CASE OF Option</code>                                                   |            |           |           |

[5]

**(c)** Decomposition has been used to design the program to help the shop manager control the stock.

Describe decomposition.

[3]

**14. M/J 25/P21/Q3**

- 3 A student has been asked to create a simple guessing game program. This program will generate a random integer value between 1 and 100. It will then repeatedly prompt the user to input an integer value until they input the randomly generated value.

The student has written a structured English description:

step 1 – randomly generate an integer value between 1 and 100 inclusive

step 2 – prompt the user to input an integer value

step 3 – output an appropriate message if the value input was too high; then repeat from **step 2**

step 4 – output an appropriate message if the value input was too low; then repeat from **step 2**

step 5 – output an appropriate message if the value input was the same value that was randomly generated; then end the program.

Write a pseudocode algorithm from this structured English description.

Assume no input validation is needed.

[8]

**15. M/J 25/P22/Q2/a**

- 2 A program to calculate the pay of employees working for a company is being designed.

(a) Stepwise refinement has been used in the design of the program.

Describe stepwise refinement.

[2]

**16. M/J 25/P22/Q3**

- 3 An algorithm is designed to generate and output two unique random integers. Each integer value is between –10 and 10 inclusive.

If both integers output are negative, a third random integer between 30 and 35 inclusive will be generated and output.

Write pseudocode for this algorithm.

[8]

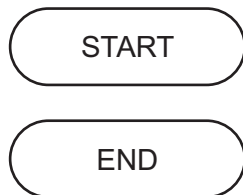
**17. M/J 25/P22/Q4**

4 An algorithm will:

- input 100 integer values, one value at a time
- store the first value input into the first location of the array `Number`
- store the next input value in the next unused location of the array `Number`
- output the contents of `Number` array in the opposite sequence to that in which the values were input.

Complete the program flowchart to represent the algorithm.

Variable declarations are **not** required.



[6]

**18. M/J 25/P23/Q3**

3 A programmer has been asked to create a module `RollDice()` to simulate multiple rolls of a dice. This module will be used as part of a program for a game.

The module will:

- step 1 – take a positive integer parameter representing the number of times the dice will be rolled
- step 2 – simulate **one** roll of the dice by generating a random integer between 1 and 6 inclusive
- step 3 – output the integer generated
- step 4 – repeat, as required from **step 2**
- step 5 – calculate the average value of the random integers generated
- step 6 – return the average value.

Write pseudocode for the module `RollDice()`

[7]

**19. M/J 25/P23/Q4/a**

- 4** A programmer is developing a new stock control program for a shop owner to produce sales reports.

**(a)** A text file `Sales.txt` stores strings representing the number of items sold.

The file contains 52 lines, each representing the number of items sold in each week of the year.

For example:

| File line number                                                                  | File data |
|-----------------------------------------------------------------------------------|-----------|
| 1                                                                                 | "350"     |
| 2                                                                                 | "502"     |
| 3                                                                                 | "434"     |
|  |           |
| 49                                                                                | "492"     |
| 50                                                                                | "872"     |
| 51                                                                                | "772"     |
| 52                                                                                | "201"     |

The example shows that 502 items were sold in week 2.

The owner requires a module to output all week numbers where the number of items sold is greater than 500.

Describe an algorithm that produces the required module.

Do **not** include pseudocode statements in your answer.

[5]

**20. O/N 24/P21/Q3/a,b**

- 3** An algorithm will output the **last** three lines from a text file `Result.txt`

The lines need to be output in the same order as they appear in the file.

Assume:

- Three variables `LineX`, `LineY` and `LineZ` will store the three lines. These are of type string and all three variables have been initialised to an empty string.
- The file exists and contains **at least** three lines.

- (a)** The algorithm to output the lines is expressed in eight steps.

Complete the steps.

1. Open the file .....
2. Loop until .....
3. .... and store in `ThisLine`
4. Assign `LineY` to `LineX`
5. Assign `LineZ` to `LineY`
6. Assign `ThisLine` to `LineZ`
7. After the loop, .....
8. Output `LineX`, `LineY`, `LineZ`

[4]

- (b)** Explain the purpose of steps 4, 5 and 6 in the algorithm from part **(a)**.

[1]

**21. O/N 24/P22/Q7/a,b**

- 7** A coffee shop runs a computerised loyalty card system.

Customers are issued with a loyalty card with their name together with a unique customer ID.

Loyalty points are added to their card each time they spend money at the shop.

The following information is stored for each customer: ID, name, home address, email address, mobile phone number, date of birth, number of points, date of last visit and amount of money spent at last visit.

A new module will generate a personalised email message to each loyalty card customer who has not visited the coffee shop in the last three months. The message will include a unique voucher code which can be used to authorise a discount if the customer goes to the shop within the next two weeks.

- (a)** Identify **three** items of customer information that will be used by the new module **and** justify your choices.

Item 1

Justification

Item 2

Justification

Item 3

Justification

[3]

- (b)** Identify the computational thinking skill that you needed to use to answer part **(a)**.

[1]

**22. O/N 24/P23/Q1/a**

- 1** **(a)** The following table contains pseudocode examples.

Each example may contain statements that relate to one or more of the following:

- selection
- iteration (repetition)
- subroutine (procedure or function).

Complete the table by placing **one or more** ticks ('✓') in each row.

| Pseudocode example                                                                         | Selection | Iteration | Subroutine |
|--------------------------------------------------------------------------------------------|-----------|-----------|------------|
| FOR Index ← 1 TO 3<br>IF Safe[Index] = TRUE THEN<br>Flap[Index] ← 0<br>ENDIF<br>NEXT Index |           |           |            |
| CASE OF Compound(3)                                                                        |           |           |            |
| REPEAT UNTIL AllDone() = TRUE                                                              |           |           |            |
| WHILE Result[3] <> FALSE                                                                   |           |           |            |

[4]

**23. O/N 24/P23/Q2/a,b**

**2** An algorithm will:

1. prompt and input a sequence of 100 integer values, one at a time
2. sum the positive integers
3. output the result of the sum.

**(a)** Write pseudocode for the algorithm.

Assume the value zero is neither positive nor negative.

You must declare all variables used in the algorithm.

[5]

**(b)** The algorithm requires the use of basic constructs. One of these is sequence.

Identify **one other** basic construct required by the algorithm **and** describe how it is used.

Construct

Use

[2]

**24. O/N 24/P23/Q3/b,c**

- 3** The implementation of a linked list uses an integer variable and a 1D array `List` of type `Node`.

Record type `Node` is declared in pseudocode as follows:

```
TYPE Node
  DECLARE Data : STRING
  DECLARE Pointer : INTEGER
ENDTYPE
```

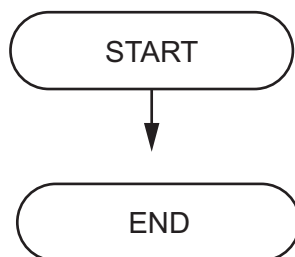
The array `List` is declared in pseudocode as follows:

```
DECLARE List : ARRAY[1:200] OF Node
```

The linked list will operate as follows:

- Integer variable `HeadPointer` will store the array index for the first node in the linked list.
  - The `Pointer` field of a node contains the index value of the next array element (the next node) in the linked list.
  - The value 0 is used as a null pointer.
- (b)** The array `List` will be initialised so that each node points to the following node. The last node will contain a null pointer.

Complete the program flowchart to represent the algorithm for this operation.



[4]

- (c)** An algorithm outputs the `Data` field from all nodes in the array `List`. The order the `Data` is output should be the same order it is stored in the linked list.

Describe the algorithm in **four** steps.

Do **not** use pseudocode statements in your answer.

Step 1

Step 2

Step 3

Step 4

[4]

**25. O/N 24/P23/Q7/b**

- 7** A coffee shop owner wants to introduce a computerised loyalty card system.

A programmer discusses the details of the system with the shop owner.

- (b)** The shop will give each customer a loyalty card that displays a unique customer ID as a bar code. A customer will be able to present their card each time they make a purchase. The system will scan the bar code, calculate points, and add them to the customer's total. When the customer next makes a purchase and presents their card, they will have the option to exchange points for a discount.

The designer decides that this activity will be handled by a new module. Decomposition will be used to break the problem of designing the new module down into sub-problems (sub-modules).

Identify **four** sub-modules that could be used in the design of the new module **and** describe their use.

Sub-module 1

Use

Sub-module 2

Use

Sub-module 3

Use

Sub-module 4

Use

[4]

**26. M/J 24/P21/Q1/c**

- 1 An algorithm is developed in pseudocode before being coded in a programming language.

(a) The following table shows four valid pseudocode assignment statements.

Complete the table by giving an appropriate data type to declare each of the variables A, B, C and D.

| Assignment statement  | Data type |
|-----------------------|-----------|
| A ← LEFT(MyName, 1)   |           |
| B ← Total * 2         |           |
| C ← INT(ItemCost) / 3 |           |
| D ← "Odd OR Even"     |           |

[4]

(c) The variable names A, B, C and D in part (a) are **not** good programming practice.

(i) State why these variable names are **not** suitable.

[1]

(ii) Identify **one** problem that these variable names might cause.

[1]

(iii) The choice of suitable variable names is one example of good programming practice.

Give **one other** example.

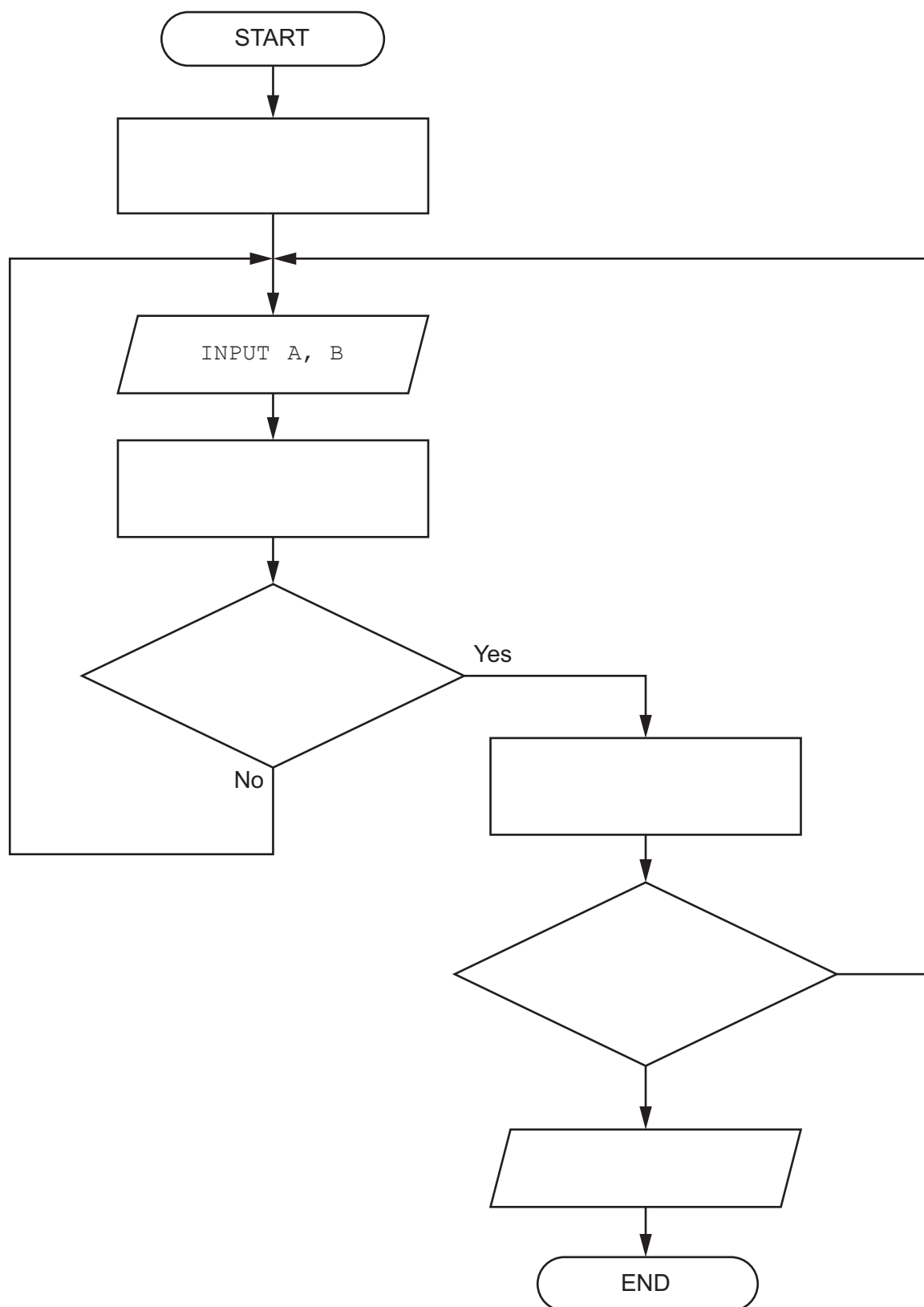
[1]

**27. M/J 24/P21/Q2/a**

- 2 An algorithm has three steps. It will:

1. repeatedly input a pair of numeric values A and B
2. count the number of pairs that are input until A has been greater than B 10 times
3. output the number of pairs that were input.

(a) Complete the program flowchart.



**28. M/J 24/P21/Q7/a**

- 7 A fitness club has a computerised membership system. The fitness club offers a number of different exercise classes.

The following information is stored for each club member: name, home address, email address, mobile phone number, date of birth and the exercise(s) they are interested in.

- (a) When an exercise class is planned, a new module will send personalised text messages to each member who has expressed an interest in that exercise. Members wishing to join the class send a text message back. Members may decide **not** to receive future text messages by replying with the message 'STOP'.

The process of abstraction is used to filter out unnecessary information.

- (i) State **one** advantage of applying abstraction to this problem.

[1]

- (ii) Identify **three** items of information that will be required by the new module. Justify your choices with reference to the given scenario.

Item 1 required

Justification

Item 2 required

Justification

Item 3 required

Justification

[3]

- (iii) Identify **two** operations that would be required to process data when the new module receives a text message back from a member.

Operation 1

Operation 2

[2]

**29. M/J 24/P22/Q1/a,c**

- 1 (a) The following table contains pseudocode examples.

Each example may contain statements that relate to one or more of the following:

- selection
- iteration (repetition)
- input/output.

Complete the table by placing **one or more** ticks (✓) in each row.

| Pseudocode example                                                         | Selection | Iteration | Input/Output |
|----------------------------------------------------------------------------|-----------|-----------|--------------|
| FOR Index $\leftarrow$ 1 TO 10<br>Data[Index] $\leftarrow$ 0<br>NEXT Index |           |           |              |
| WRITEFILE ThisFile, "*****"                                                |           |           |              |
| UNTIL Level > 25                                                           |           |           |              |
| IF Mark > 74 THEN<br>READFILE OldFile, Data<br>ENDIF                       |           |           |              |

[4]

(b) Program variables have data types as follows:

| Variable | Data type |
|----------|-----------|
| MyChar   | CHAR      |
| MyString | STRING    |
| MyInt    | INTEGER   |

Complete the table by filling in each gap with a function (from the **insert**) so that each expression is valid.

| Expression                                      |
|-------------------------------------------------|
| MyInt $\leftarrow$ ..... (3.1415926)            |
| MyChar $\leftarrow$ ..... ("Elwood", 3, 1)      |
| MyString $\leftarrow$ ..... ( ..... (27.509))   |
| MyInt $\leftarrow$ ..... ( ..... ("ABC123", 3)) |

[4]

(c) The variables given in part (b) are chosen during the design stage of the program development life cycle.

The choices are to be documented to simplify program maintenance.

State a suitable way of documenting the variables **and** give **one** piece of information that should be recorded, in addition to the data type.

[2]

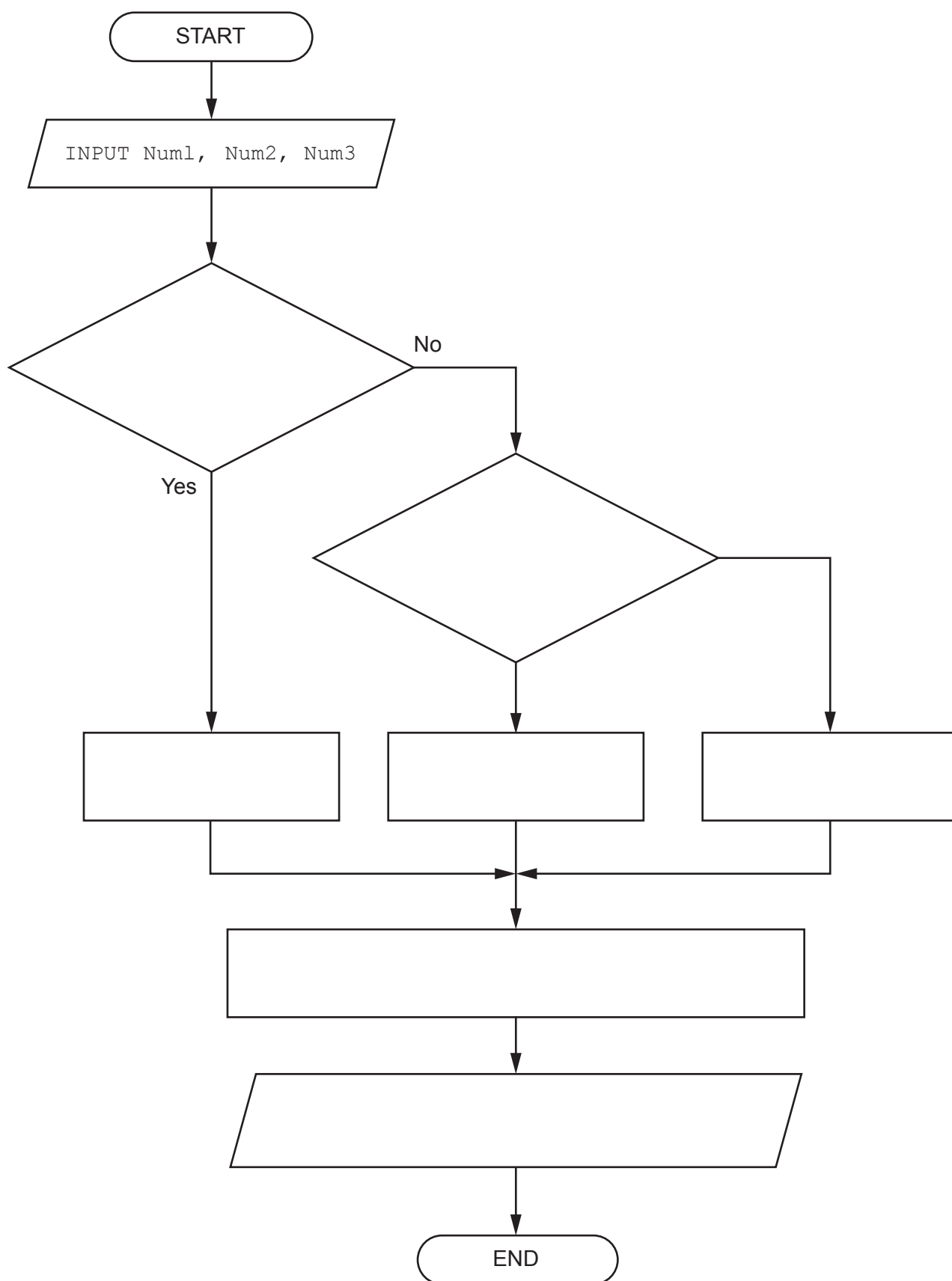
### 30. M/J 24/P22/Q2/a,b

2 A program is being developed.

- (a) An algorithm for part of the program will:
- input three numeric values and assign them to identifiers `Num1`, `Num2` and `Num3`
  - assign the largest value to variable `Ans`
  - output a message giving the largest value and the average of the three numeric values.

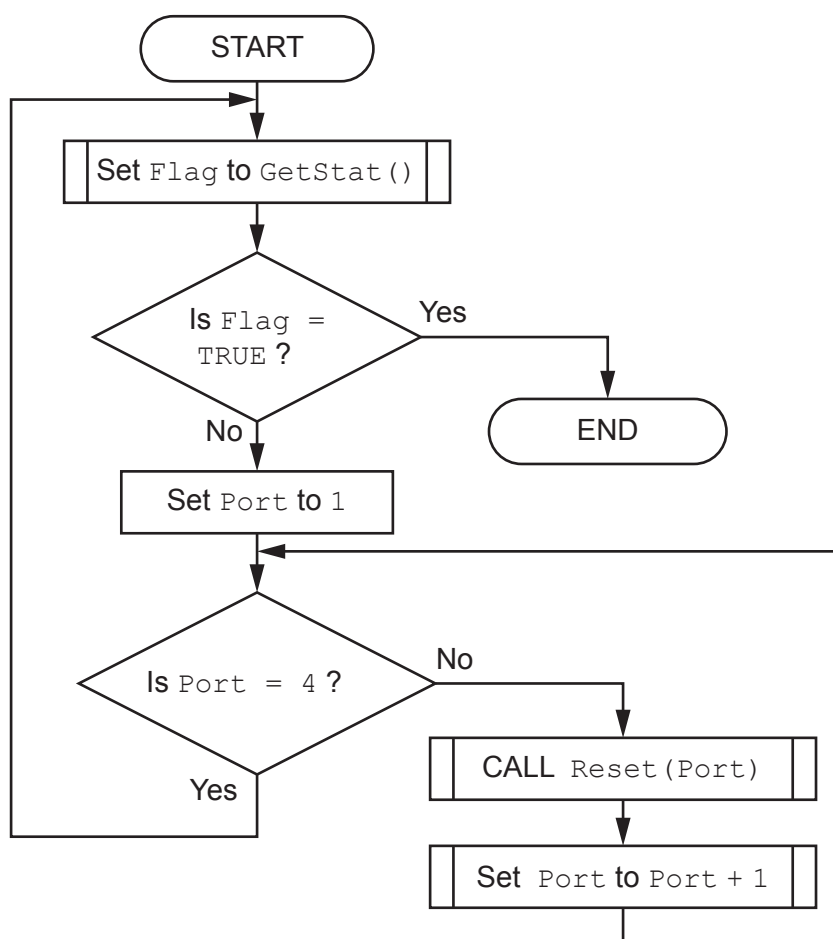
Assume the values are all different and are input in no particular order.

Complete the program flowchart on page 5 to represent the algorithm.



[5]

(b) A different part of the program contains an algorithm represented by the following program flowchart:



Write pseudocode for the algorithm.

[5]

**31. M/J 24/P22/Q7/a**

- 7** A fitness club has a computerised membership system.

The system stores information for each club member: name, home address, email address, mobile phone number, date of birth and exercise preferences.

Many classes are full, and the club creates a waiting list for each class. The club adds details of members who want to join a class that is full to the waiting list for that class.

When the system identifies that a space is available in one of the classes, a new module will send a text message to each member who is on the waiting list.

- (a)** Decomposition will be used to break the new module into sub-modules (sub-problems).

Identify **three** sub-modules that could be used in the design **and** describe their use.

Sub-module 1

Use

Sub-module 2

Use

Sub-module 3

Use

[3]

**32. M/J 24/P22/Q8/a**

- 8** A teacher is designing a program to process pseudocode projects written by her students.

Each student project is stored in a text file.

The process is split into a number of stages. Each stage performs a different task and creates a new file.

For example:

| File name           | Comment                                               |
|---------------------|-------------------------------------------------------|
| MichaelAday_src.txt | Student project file produced by student Michael Aday |
| MichaelAday_S1.txt  | File produced by stage 1                              |
| MichaelAday_S2.txt  | File produced by stage 2                              |

- (a)** Suggest a reason why the teacher's program has been split into a number of stages **and** give the benefit of producing a different file from each stage.

Reason

Benefit

[2]

**33. M/J 24/P23/Q2/a,b**

- 2 (a) A program uses a global 1D array of type string and a text file.

An algorithm that forms part of the program is expressed as follows:

- copy the first line from the file into the first element of the array
- copy the second line from the file into the second element of the array
- continue until all lines in the file have been copied into the array.

Stepwise refinement is applied to the algorithm.

Outline **five** steps for this algorithm that could be used to produce pseudocode.

Assume there are more elements in the array than lines in the file.

Do **not** use pseudocode statements in your answer.

Step 1

Step 2

Step 3

Step 4

Step 5

[5]

- (b) Sequence is one programming construct.

Identify **one other** programming construct that will be required when the algorithm from part (a) is converted into pseudocode **and** explain its use.

Construct

Use

[2]

**34. M/J 24/P23/Q3/b**

- 3 A record structure is declared to hold data relating to components being produced in a factory:

```
TYPE Component
  DECLARE Item_ID : STRING
  DECLARE Reject : BOOLEAN
  DECLARE Weight : REAL
ENDTYPE
```

The factory normally produces a batch (or set) of 1000 components at a time. A global array is declared to store 1000 records for a batch:

```
DECLARE Batch : ARRAY [1:1000] OF Component
```

Two global variables contain the minimum and maximum acceptable weight for each component. The values represent an inclusive range and are declared as:

```
DECLARE Min, Max : REAL
```

(b) A module `InRange()` will:

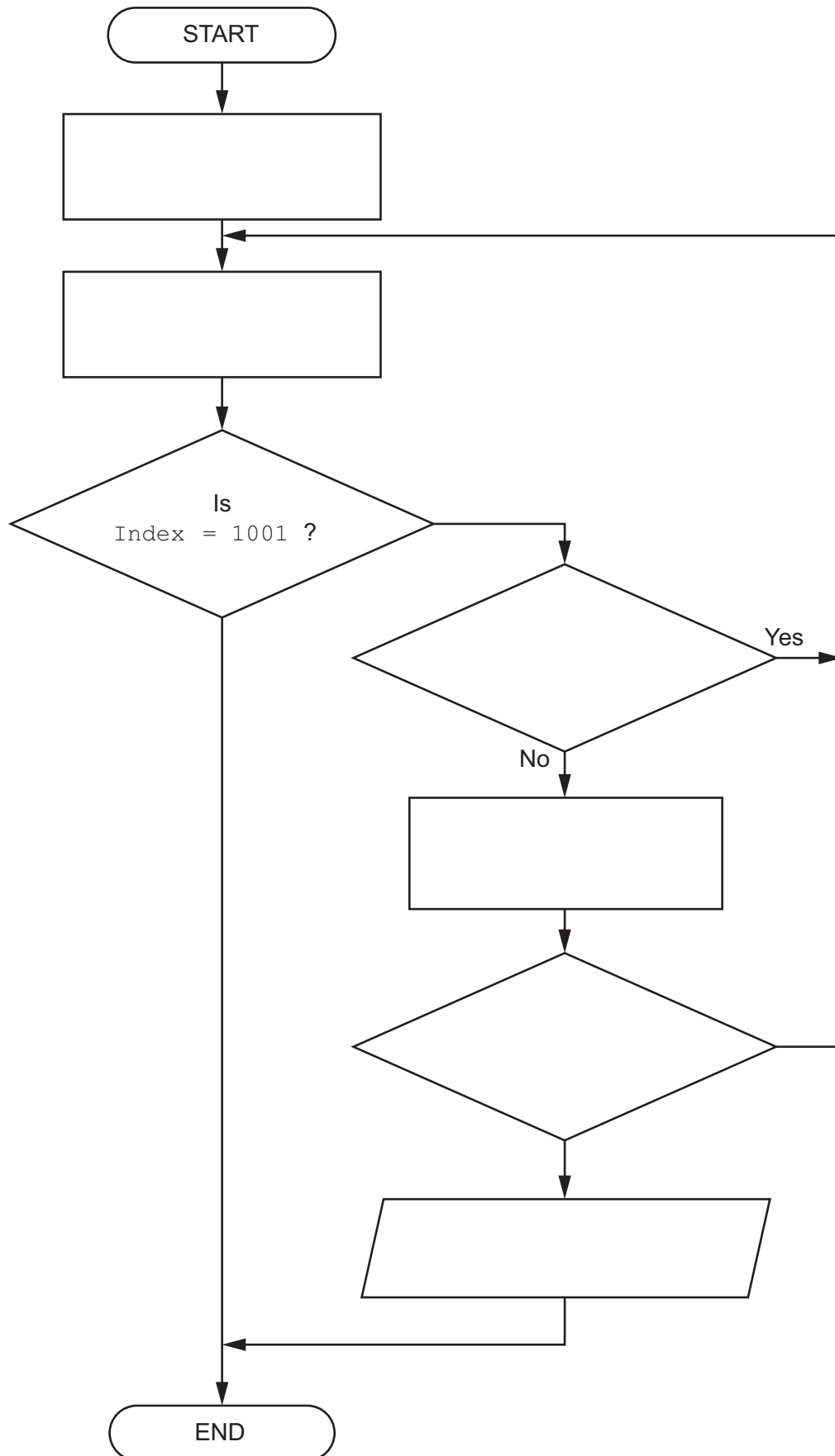
- be called with an integer parameter representing an index value of a record in the `Batch` array
- check if the weight of the indexed component is within the acceptable range
- return `TRUE` if the weight is in the range and `FALSE` if it is **not**.

A module `BatchCheck()` will:

- iterate through a batch of 1000 component records
- call module `InRange()` to check each individual component record
- keep a count of the number of components that fail
- output a suitable warning message and immediately stop if the number of failed components exceeds 5.



Complete the program flowchart to represent the algorithm for module `BatchCheck()`.



**35. O/N 23/P21/Q2/a,b**

**2 (a)** An algorithm is expressed as follows:

- input 100 numbers, one at a time
- keep a total of all numbers input that have a value between 30 and 70 inclusive and output this total after the last number has been input.

Outline, using stepwise refinement, the five steps for this algorithm which could be used to produce pseudocode.

Do **not** use pseudocode statements in your answer.

Step 1

Step 2

Step 3

Step 4

Step 5

[5]

**(b)** Sequence is one programming construct.

Identify **two other** programming constructs that will be required when the algorithm is converted into pseudocode.

Construct 1

Construct 2

[2]

**36. O/N 23/P22/Q1/a**

- 1 A shop sells car accessories. A customer order is created if an item cannot be supplied from current stock. A program is being developed to create and manage the customer orders.

(a) The following identifier table shows some of the data that will be stored for each order.

Complete the identifier table by adding meaningful variable names and appropriate data types.

| Example value | Explanation                                | Variable name | Data type |
|---------------|--------------------------------------------|---------------|-----------|
| "Mr Khan"     | The name of the customer                   |               |           |
| 3             | The number of items in the order           |               |           |
| TRUE          | To indicate whether this is a new customer |               |           |
| 15.75         | The deposit paid by the customer           |               |           |

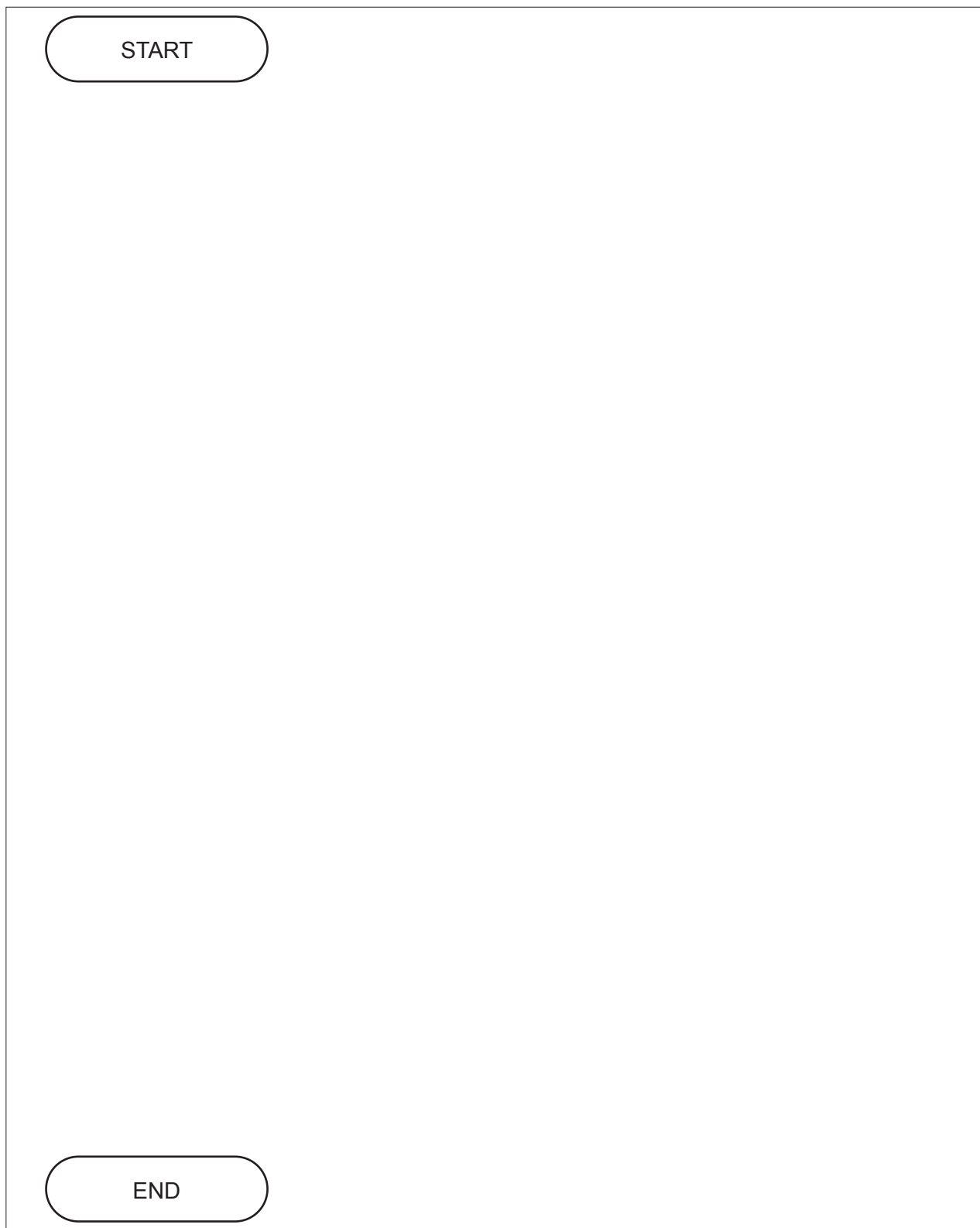
[4]

**37. O/N 23/P22/Q2/a**

- 2 An algorithm will:

1. input a sequence of integer values, one at a time
2. ignore all values until the value 27 is input, then sum the remaining values in the sequence
3. stop summing values when the value 0 is input and then output the sum of the values.

(a) Draw a program flowchart to represent the algorithm.



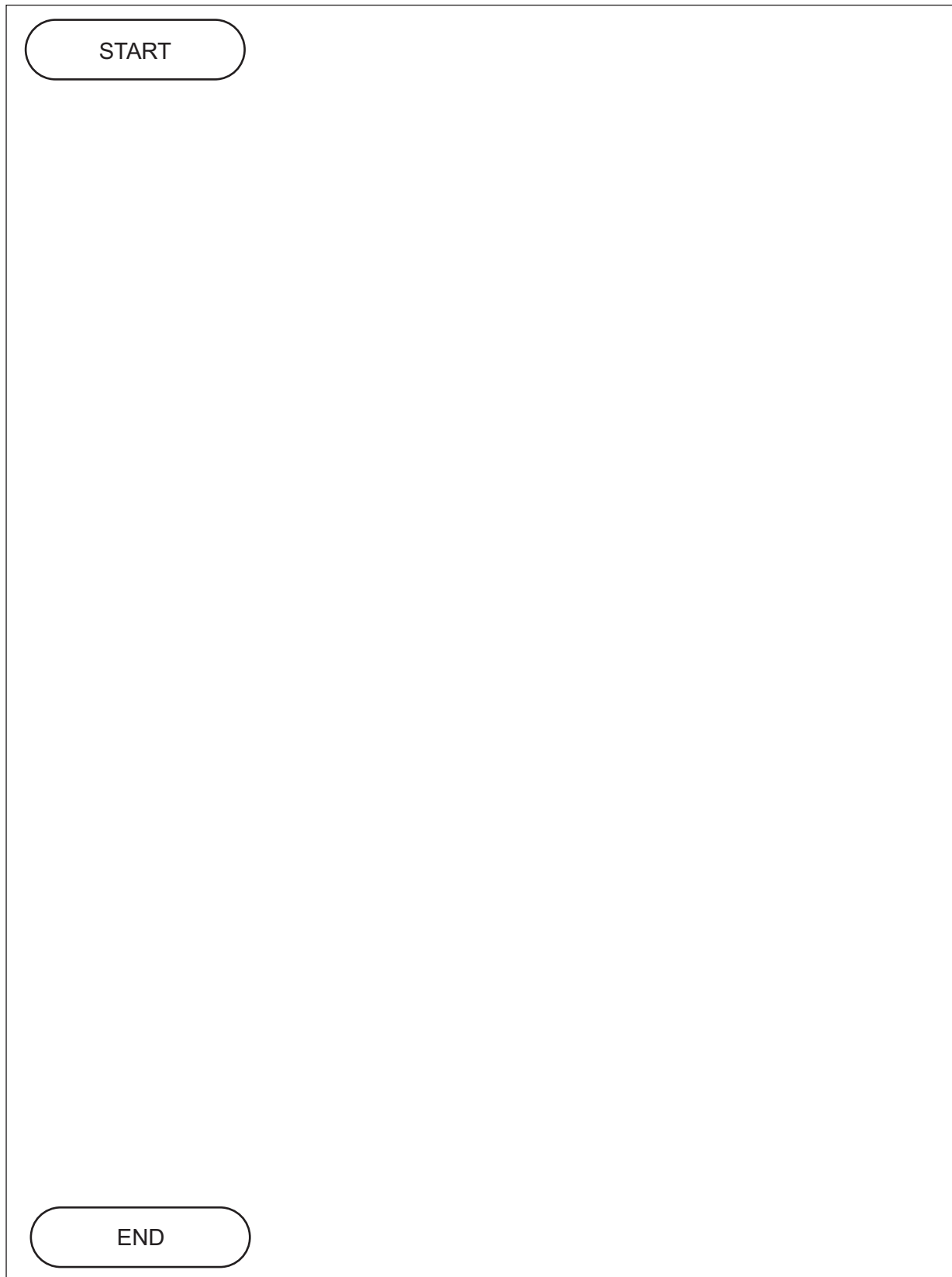
[5]

**38. O/N 23/P23/Q2/a**

**2** Data is a 1D array of integers, containing 30 elements. All element values are unique.

**(a)** An algorithm will output the index of the element with the smallest value.

Draw a program flowchart to represent the algorithm.



[5]

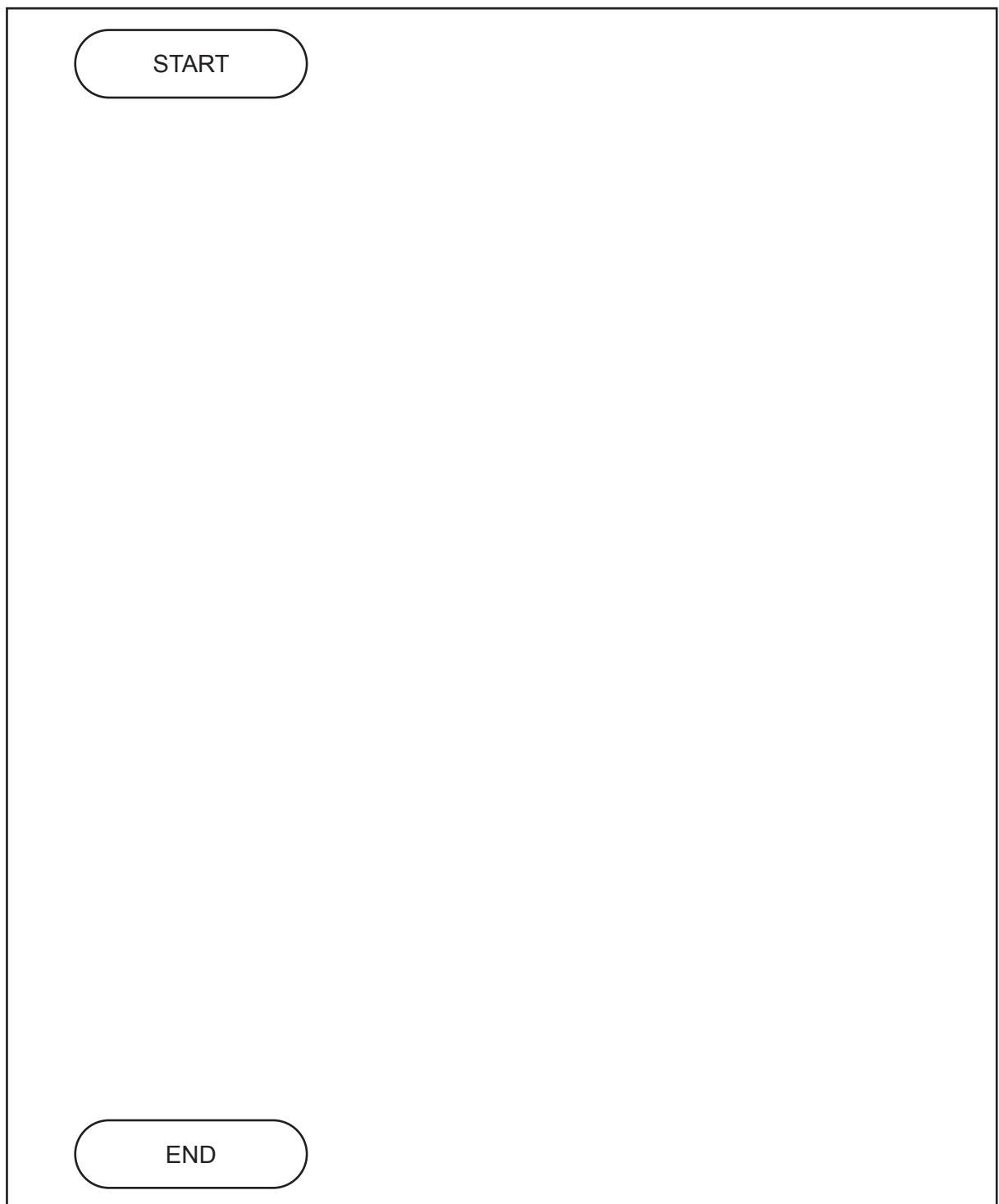
**39. M/J 23/P21/Q2/a**

- 2 A program contains an algorithm to output a string of a specified length containing identical characters.

(a) The algorithm is described as follows:

1. prompt and input a character and store in `MyChar`
2. prompt and input an integer and store in `MyCount`
3. generate a string consisting of `MyChar` repeated `MyCount` times
4. output the string.

Draw a program flowchart to represent the algorithm.



**40. M/J 23/P21/Q3/b**

- 3** Customers collect points every time they make a purchase at a store.

A program is used to manage the points system and the table lists some of the information stored for one customer.

| Information                | Data type required |
|----------------------------|--------------------|
| Name                       | String             |
| Number of points collected | Integer            |
| Date of birth              | Date               |

- (b)** Customers receive points depending on the amount they spend. The number of points depends on the band that the amount falls into:

| Band | Amount                           | Points                   |
|------|----------------------------------|--------------------------|
| 1    | Less than \$10                   | 5 per whole dollar (\$)  |
| 2    | Between \$10 and \$100 inclusive | 7 per whole dollar (\$)  |
| 3    | Over \$100                       | 10 per whole dollar (\$) |

For example, if the amount is \$99.77, this amount is in band 2 and therefore the number of points is  $7 \times 99$ , which is 693 points.

The algorithm to calculate the points from a given amount is expressed as follows:

- work out the appropriate band
- calculate and output the number of points.

Apply the process of stepwise refinement to increase the detail of the algorithm. Structure your algorithm into a sequence of five steps that could be used to produce pseudocode.

Write the **five** steps.

- 1
- 2
- 3
- 4
- 5

**41. M/J 23/P21/Q7/a**

- 7 A school has a computerised library system that allows students to borrow books for a fixed length of time. The system uses text files to store details of students, books and loans.

A new module is to be written which will generate emails to each student who has an overdue book.

- (a) Decomposition will be used to break down the problem of designing the new module into sub-problems.

Identify **three** program modules that could be used in the design **and** describe their use.

Module 1

Use

Module 2

Use

Module 3

Use

[3]

**42. M/J 23/P22/Q1/b**

- 1 A program calculates the postal cost based on the weight of the item and its destination. Calculations occur at various points in the program and these result in the choice of several possible postal costs. The programmer has built these postal costs into the program.

For example, the postal cost of \$3.75 is used in the following lines of pseudocode:

```
IF Weight < 250 AND ValidAddress = TRUE THEN
    ItemPostalCost ← 3.75      // set postal cost for item to $3.75
    ItemStatus ← "Valid"      // item can be sent
ENDIF
```

- (b) The lines of pseudocode contain features that make them easier to understand.

State **three** of these features.

1

2

3

[3]

**43. M/J 23/P22/Q2/c**

**2** A program stores a user's date of birth using a variable `MyDOB` of type `DATE`.

**(c)** The program will output the day of the week corresponding to `MyDOB`.

For example, given the date 22/06/2023, the program will output "Thursday".

An algorithm is required. An array will be used to store the names of the days of the week.

Define the array **and** describe the algorithm in **four** steps.

Do **not** use pseudocode statements in your answer.

Array definition

Step 1

Step 2

Step 3

Step 4

[6]

**44. M/J 23/P22/Q3/b**

**3** A program stores data in a text file. When data is read from the file, it is placed in a queue.

**(b)** The design of the queue is completed and the number of locations is increased.

A function `AddToQueue()` has been written. It takes a string as a parameter and adds this to the queue. The function will return `TRUE` if the string was added successfully.

A procedure `FileToQueue()` will add each line from the file to the queue. This procedure will end when all lines have been added or when the queue is full.

Describe the algorithm for procedure `FileToQueue()`.

Do **not** use pseudocode in your answer.

[5]

**45. M/J 23/P22/Q7/a**

- 7** A computer system for a shop stores information about each customer. The items of information include name and address (both postal and email) together with payment details and order history. The system also stores the product categories they are interested in and how they would like to be contacted.

- (a)** The shop wants to add a program module that will generate emails to be sent to customers who may be interested in receiving details of new products.

- (i)** State **three** items of information that the new module would need. Justify your choice in each case.

Information

Justification

Information

Justification

Information

Justification

[3]

- (ii)** Identify **two** items of customer information that would **not** be required by the new module. Justify your choice in each case.

Information

Justification

Information

Justification

[2]

**46. M/J 23/P23/Q1/d**

- 1 The following pseudocode represents part of the algorithm for a program.

Line numbers are for reference only.

```

10  DECLARE Sheet4 : ARRAY[1:2, 1:50] OF INTEGER
...
100 FOR PCount ← 0 TO 49
101     Sheet4[1, PCount] ← 0
102     Sheet4[2, PCount] ← 47
103 NEXT PCount

```

- (d) The pseudocode statements given in the following table are used in other parts of the algorithm.

Complete the table by placing **one or more** ticks (✓) in each row.

The first row has already been completed.

| Pseudocode statement         | Input | Process | Output |
|------------------------------|-------|---------|--------|
| INPUT MyChoice               | ✓     |         |        |
| OUTPUT FirstName & LastName  |       |         |        |
| WRITEFILE YourFile, TextLine |       |         |        |
| READFILE MyFile, TextLine    |       |         |        |
| Result ← SQRT (NextNum)      |       |         |        |

[4]

**47. M/J 23/P23/Q3/b(ii)**

- 3 A program processes data using a stack. The data is copied to a text file before the program ends.

- (b) The data is copied to a text file before the program ends.

- (ii) A module `SaveStack()` will write the data from the stack to a text file.

Express an algorithm for `SaveStack()` as five steps that could be used to produce pseudocode.

Write the **five** steps.

Step 1

Step 2

Step 3

Step 4

Step 5

[5]

**48. M/J 23/P23/Q7/a**

- 7 A school has a library system which allows students to borrow books for a length of time. Information relating to students and books is stored in text files. Student information includes name, home address, email address, date of birth, tutor and subject choices. Book information includes author, title, subject category, library location and the date that the book was borrowed.

A program helps the staff to manage the borrowing of books.

- (a) A new module needs to be written to generate emails to send to students who have an overdue book. Students who are sent an email are prevented from borrowing any more books until the overdue book is returned.

The process of abstraction has been used when designing the new module.

- (i) State the purpose of applying abstraction to this problem.

[1]

- (ii) Identify **one** item of information that is required and **one** item that is **not** required in the new module. Justify your choices.

Item required

Justification

Item not required

Justification

[2]

- (iii) Identify **two** operations that would be required to process data when an overdue book is returned.

Operation 1

Operation 2

[2]

**49. O/N 22/P21/Q2/a,b**

- 2** A system is being developed to help manage a car hire business. A customer may hire a car for a number of days.

An abstract model needs to be produced.

- (a)** Explain the process of abstraction **and** state **four** items of data that should be stored each time a car is hired.

Explanation

Item 1

Item 2

Item 3

Item 4

[3]

- (b)** Identify **two** operations that would be required to process the car hire data.

Operation 1

Operation 2

[2]

**50. O/N 22/P21/Q3**

- 3** A 1D array `Data` of type integer contains 200 elements. Each element has a unique value.

An algorithm is required to search for the largest value and output it.

Describe the steps that the algorithm should perform.

Do **not** include pseudocode statements in your answer.

[5]

**51. O/N 22/P22/Q2/a**

also Unit 10 (2 marks); this unit 5 marks

- 2 (a)** An algorithm will process data from a test taken by a group of students. The algorithm will prompt and input the name and test mark for each of the 35 students.

The algorithm will add the names of all the students with a test mark of less than 20 to an existing text file `Support_List.txt`, which already contains data from other group tests.

- (i) Describe the steps that the algorithm should perform.

Do **not** include pseudocode statements in your answer.

[5]

- (ii) Explain why it may be better to store the names of the students in a file rather than in an array.

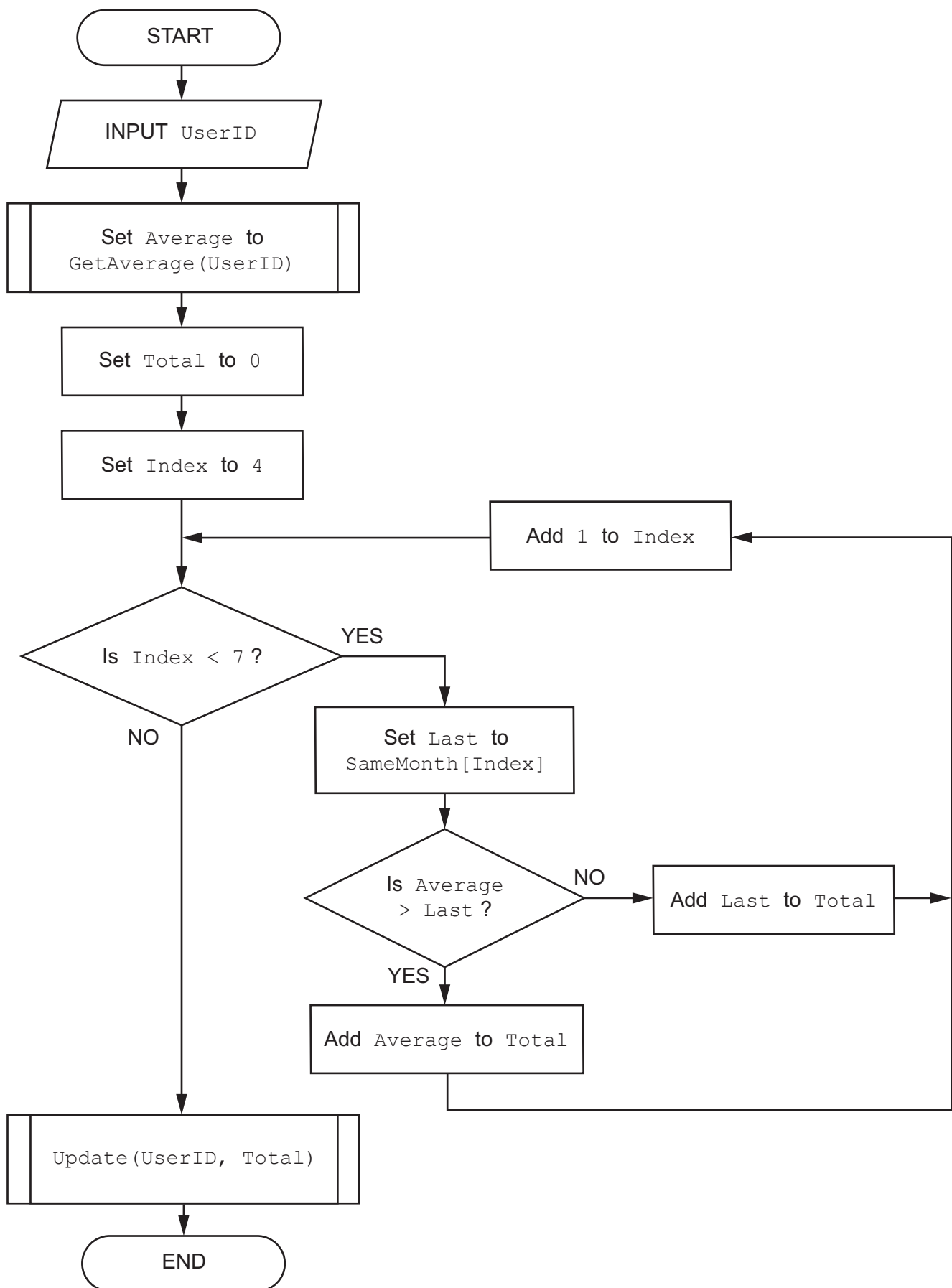
[1]

- (iii) Explain why `WRITE` mode cannot be used in the answer to **part 2(a)(i)**.

[1]

**52. O/N 22/P22/Q4/a**

- 4** The program flowchart represents a simple algorithm.



(a) Write the equivalent pseudocode for the algorithm represented by the flowchart.

**53. O/N 22/P22/Q5**

**5** Examine the following pseudocode.

```
IF A = TRUE THEN
  IF B = TRUE THEN
    IF C = TRUE THEN
      CALL Sub1 ()
    ELSE
      CALL Sub2 ()
    ENDIF
  ENDIF
ELSE
  IF B = TRUE THEN
    IF C = TRUE THEN
      CALL Sub4 ()
    ELSE
      CALL Sub3 ()
    ENDIF
  ELSE
    IF C = FALSE THEN
      CALL Sub3 ()
    ELSE
      CALL Sub4 ()
    ENDIF
  ENDIF
ENDIF
```

A programmer wants to re-write the pseudocode as **four** separate IF...THEN...ENDIF statements, each containing a single CALL statement. This involves writing a single, simplified logic expression as the condition in each statement.

Write the amended pseudocode.

- 1
- 2
- 3
- 4

[4]

**54. O/N 22/P23/Q2**

- 2 A program is being designed for a smartphone to allow users to send money to the charity of their choice.

Decomposition will be used to break the problem down into sub-problems.

Identify **three** program modules that could be used in the design **and** describe their use.

Module 1

Use

Module 2

Use

Module 3

Use

[3]

**55. O/N 22/P23/Q3**

- 3 An algorithm is needed to process a sequence of numbers. Numbers may be positive or negative and may be integer or decimal.

The algorithm will:

- prompt and input one number at a time until the value zero is input
- sum the negative numbers
- sum the positive numbers
- when zero is input, output the two sum values and then end.

Describe the algorithm needed. Do **not** include pseudocode statements in your answer.

[5]

**56. O/N 22/P23/Q5/c**

- 5 (a) A text string contains three data items concatenated as shown:

<StockID><Description><Cost>

Item lengths are:

| Item        | Length                      |
|-------------|-----------------------------|
| StockID     | 5                           |
| Description | 32                          |
| Cost        | the remainder of the string |

A procedure `Unpack()` takes four parameters of type string. One parameter is the original text string. The other three parameters are used to represent the three data items shown in the table and are assigned values within the procedure. These values will be used by the calling program after the procedure ends.

- (c) `Unpack()` is part of a program made up of several modules. During the design stage, it is important to follow good programming practice. One example of good practice is the use of meaningful identifier names.

Give the reason why this is good practice. Give **two other** examples of good practice.

Reason

Example 1

Example 2

[3]

**57. O/N 22/P23/Q7/b(ii)**

- 7 A teacher is designing a program to perform simple syntax checks on programs written by students.

Two global 1D arrays are used to store the syntax error data. Both arrays contain 500 elements.

- Array `ErrCode` contains integer values that represent an error number in the range 1 to 800.
- Array `ErrMsgText` contains string values that represent an error description.

The following diagram shows an example of the arrays.

| Index | ErrCode                                                                             | ErrMsgText                                                                           |
|-------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 1     | 10                                                                                  | "Invalid identifier name"                                                            |
| 2     | 20                                                                                  | "Bracket mismatch"                                                                   |
| 3     | 50                                                                                  | ""                                                                                   |
| 4     | 60                                                                                  | "Type mismatch in assignment"                                                        |
| ...   |  |  |
| 500   | 999                                                                                 | <Undefined>                                                                          |

**Note:**

- There are less than 500 error codes so corresponding elements in both arrays may be unused. Unused elements in `ErrCode` have the value 999. These will occur at the end of the array. The value of unused elements in `ErrText` is undefined.
- Values in the `ErrCode` array are stored in ascending order but not all values may be present. For example, there may be no error code 31.
- Some error numbers are undefined. In these instances, the `ErrCode` array will contain a valid error number but the corresponding `ErrText` element will contain an empty string.

The teacher has defined one program module as follows:

| Module                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>OutputRange()</code> | <ul style="list-style-type: none"> <li>• Prompts for input of two error numbers</li> <li>• Outputs a list of error numbers between the two numbers input (inclusive) together with the corresponding error description</li> <li>• Outputs a warning message when the error description is missing as for error number 50 in the example</li> <li>• Outputs a suitable header and a final count of error numbers found</li> </ul> <p>Output based on the example array data above:</p> <pre>List of error numbers from 1 to 60  10 : Invalid identifier name 20 : Bracket mismatch 50 : Error Text Missing 60 : Type mismatch in assignment  4 error numbers output</pre> |

- (ii) A new Module `RemoveError()` will remove a given error number from the array.

Describe the algorithm that would be required. Do **not** include pseudocode statements in your answer.

**58. O/N 21/P21/Q1/c**

- 1 Sylvia is testing a program that has been written by her colleague. Her colleague tells her that the program does not contain any syntax errors.
- (c) An airline wants to provide passengers with information about individual flights and allow them to book their flight using an online booking system.
- (i) Tick (✓) **one** box in each row of the table to indicate whether each item of information would be essential for the customer when making the booking.

| Information                 | Essential | Not essential |
|-----------------------------|-----------|---------------|
| Departure time              |           |               |
| Flight number               |           |               |
| Departure airport           |           |               |
| Aircraft type               |           |               |
| Ticket price                |           |               |
| Number of seats in aircraft |           |               |

[3]

- (ii) Identify the technique used to filter out information that is not essential when designing the booking system **and** state one benefit of this technique.

Technique

Benefit

[2]

- (iii) Identify **two additional** pieces of essential information that a passenger might need when booking a flight.

1

2

[2]

**59. O/N 21/P21/Q2/a,b**

- 2 (a) An algorithm to sort a 1D array into ascending order is described as follows:

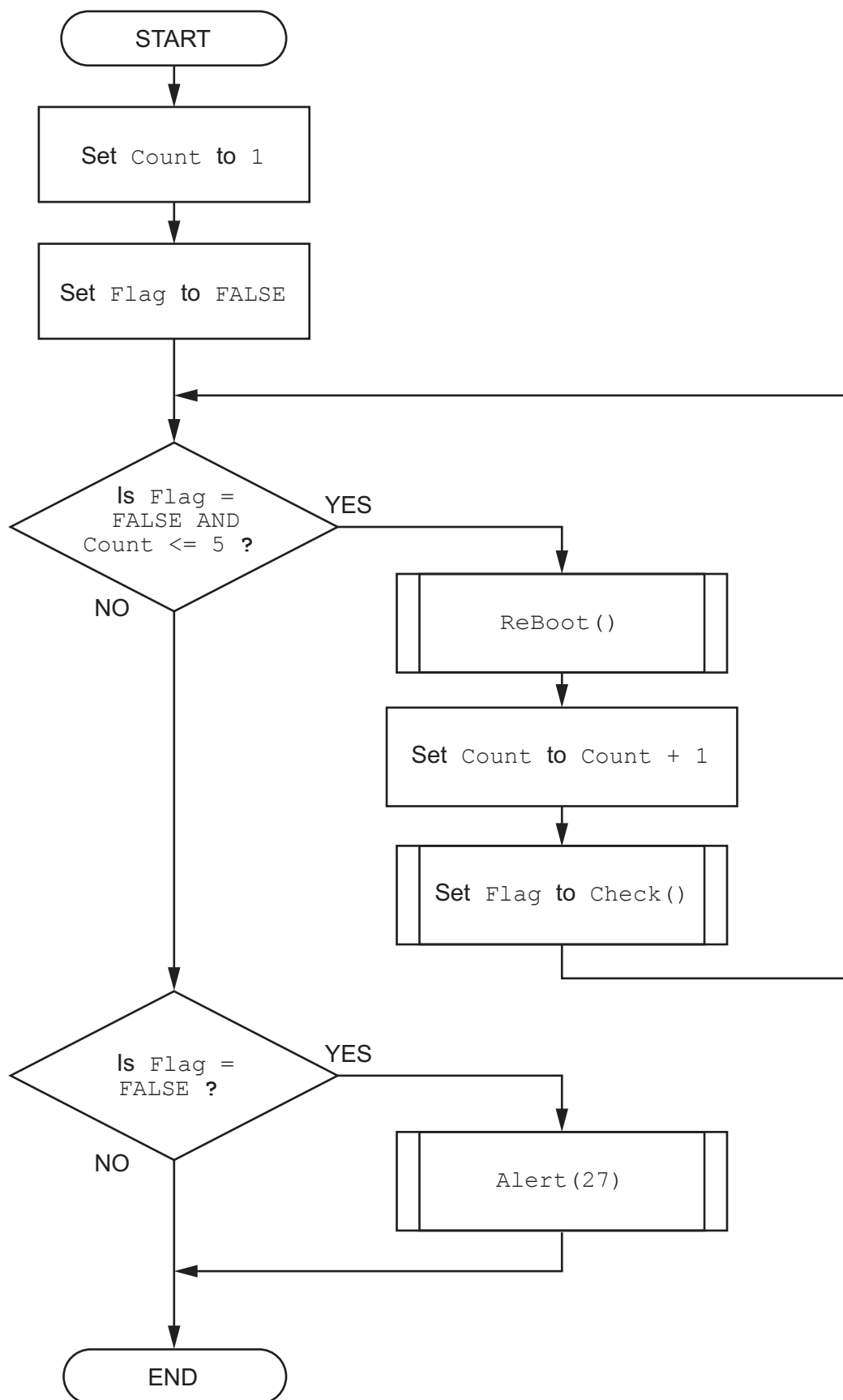
- move the largest value to the end
- keep repeating until the array is sorted.

Apply the process of stepwise refinement to this algorithm in order to produce a more detailed description.

Write the more detailed description using **structured English**. Your explanation of the algorithm should not include pseudocode statements.

[6]

- (b) The program flowchart shown describes a simple algorithm.



Write pseudocode for the simple algorithm shown on page 6.

**60. O/N 21/P22/Q1/a**

- 1 (a) A programmer applies decomposition to a problem that she has been asked to solve.

Describe decomposition.

[2]

**61. O/N 21/P22/Q2/a**

- 2 (a) An algorithm will:

1. input an integer value
2. jump to step 6 if the value is zero
3. sum and count the positive values
4. sum and count the negative values
5. repeat from step 1
6. output the two sum values and the two count values.

Draw a program flowchart on the following page to represent the algorithm.

Note that variable declarations are not required in program flowcharts.



**62. O/N 21/P22/Q4/b**

- 4** The following is a procedure design in pseudocode.

Line numbers are given for reference only.

```
10  PROCEDURE Check(InString : STRING)
11      DECLARE Odds, Evens, Index : INTEGER
12
13      Odds ← 0
14      Evens ← 0
15      Index ← 1
16
17      WHILE Index <= LENGTH(InString)
18          IF STR_TO_NUM(MID(InString, Index, 1)) MOD 2 <> 0 THEN
19              Odds ← Odds + 1
20          ELSE
21              Evens ← Evens + 1
22          ENDIF
23          Index ← Index + 1
24      ENDWHILE
25
26      CALL Result(Odds, Evens)
27  ENDPROCEDURE
```

- (b)** The pseudocode includes several features that make it easier to read and understand.

Identify **three** of these features.

1

2

3

[3]

**63. O/N 21/P23/Q1/c**

- 1 Sylvia is testing a program that has been written by her colleague. Her colleague tells her that the program does not contain any syntax errors.
- (c) An airline wants to provide passengers with information about individual flights and allow them to book their flight using an online booking system.
- (i) Tick (✓) **one** box in each row of the table to indicate whether each item of information would be essential for the customer when making the booking.

| Information                 | Essential | Not essential |
|-----------------------------|-----------|---------------|
| Departure time              |           |               |
| Flight number               |           |               |
| Departure airport           |           |               |
| Aircraft type               |           |               |
| Ticket price                |           |               |
| Number of seats in aircraft |           |               |

[3]

- (ii) Identify the technique used to filter out information that is not essential when designing the booking system **and** state one benefit of this technique.

Technique

Benefit

[2]

- (iii) Identify **two additional** pieces of essential information that a passenger might need when booking a flight.

1

2

[2]

**64. O/N 21/P23/Q2/a,b**

- 2 (a) An algorithm to sort a 1D array into ascending order is described as follows:

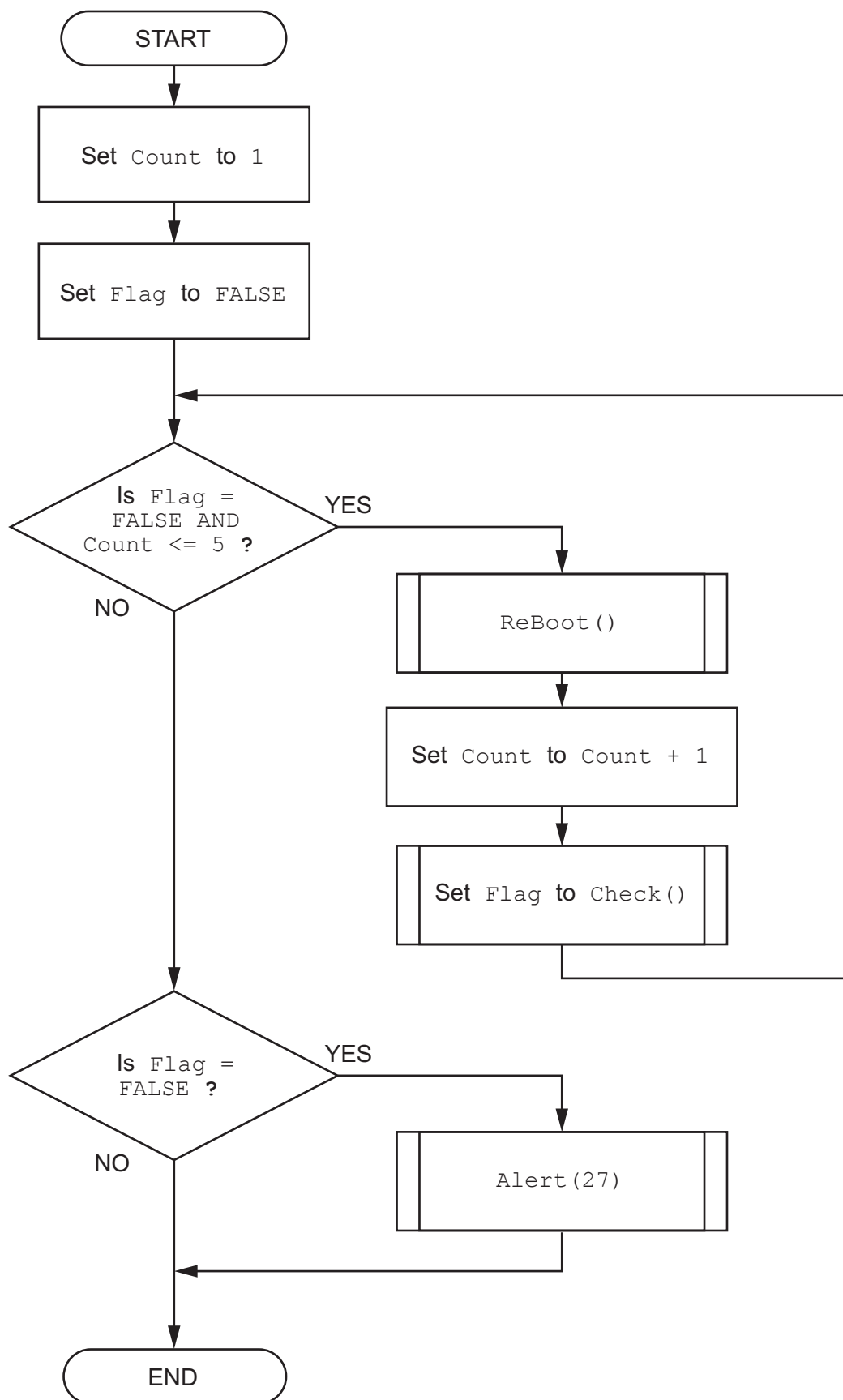
- move the largest value to the end
- keep repeating until the array is sorted.

Apply the process of stepwise refinement to this algorithm in order to produce a more detailed description.

Write the more detailed description using **structured English**. Your explanation of the algorithm should not include pseudocode statements.

[6]

- (b) The program flowchart shown describes a simple algorithm.



Write pseudocode for the simple algorithm shown on page 6.

**65. M/J 21/P21/Q1/a**

- 1 (a) A program is being developed to help manage the membership of a football club.

Complete the following identifier table.

| Example value | Explanation                                                                                         | Variable name | Data type |
|---------------|-----------------------------------------------------------------------------------------------------|---------------|-----------|
| "Wong"        | The preferred name of the member joining the football club                                          |               |           |
| FALSE         | A value to indicate whether an existing member of the club lives at the same address                |               |           |
| 19/02/1983    | When the member joined the football club                                                            |               |           |
| 1345          | The number of points a member has earned. Members of the club earn points for different activities. |               |           |

[4]

**66. M/J 21/P21/Q2/c**

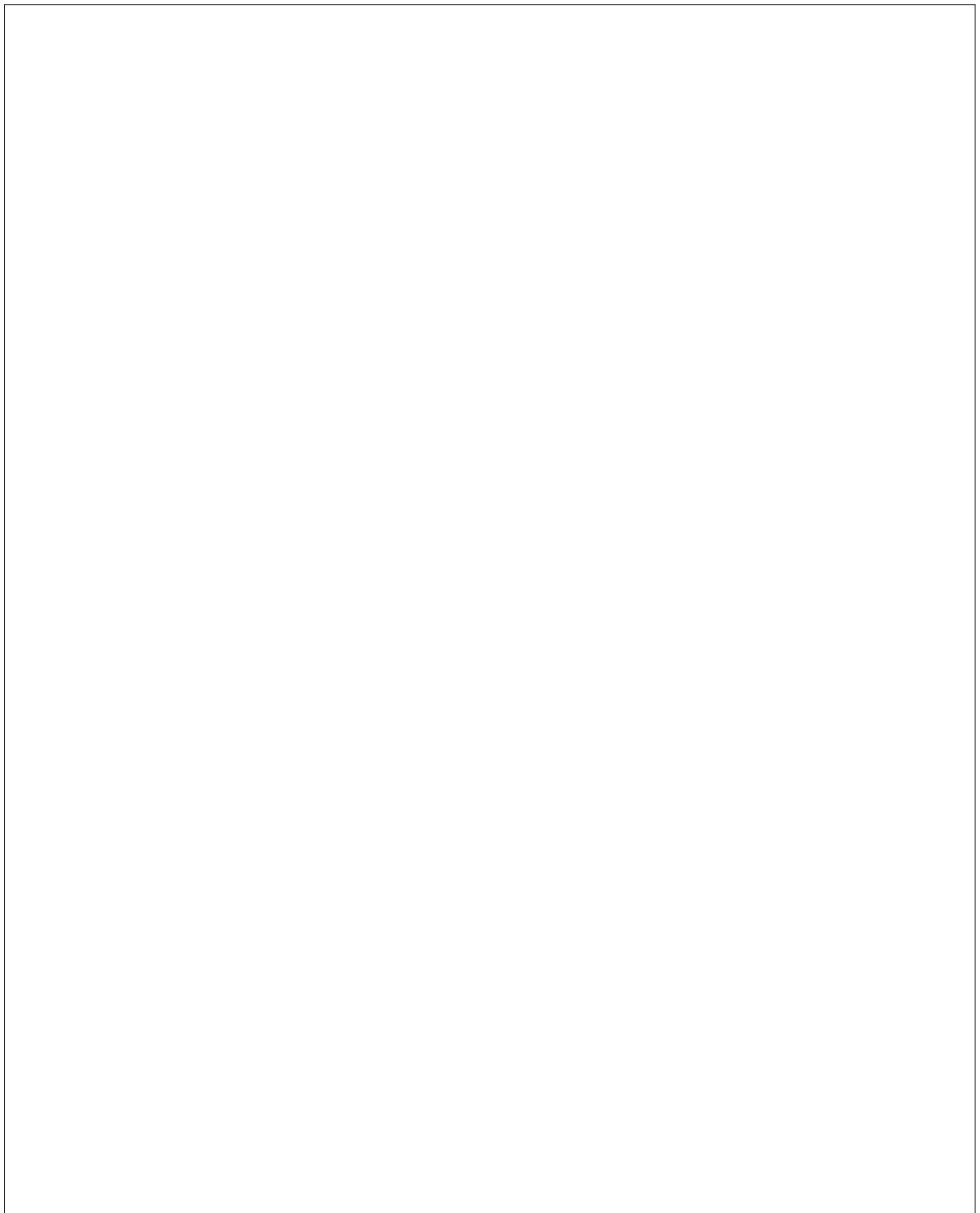
- (c) A program will:

- input 50 unique integer values
- output the largest value
- output the average of the values **excluding** the largest value.

Draw a program flowchart to represent the algorithm.

Variable declarations are **not** required.

It is not necessary to check that each input value is unique.



[6]

**67. M/J 21/P21/Q7/b**

**7** A program is needed to take a string containing a full name and produce a new string of initials.

Some words in the full name will be ignored. For example, "the", "and", "of", "for" and "to" may all be ignored.

Each letter of the abbreviated string must be upper case.

For example:

| Full name                                              | Initials |
|--------------------------------------------------------|----------|
| Integrated Development Environment                     | IDE      |
| The American Standard Code for Information Interchange | ASCII    |

The programmer has decided to use a global variable `FNString` of type `STRING` to store the full name.

It is assumed that:

- words in the full name string are separated by a single space character
- space characters will not occur at the beginning or the end of the full name string
- the full name string contains at least one word.

The programmer has started to define program modules as follows:

| Module                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>GetStart()</code> | <ul style="list-style-type: none"> <li>• Called with an <code>INTEGER</code> as a parameter, representing the number of a word in <code>FNString</code>.</li> <li>• Returns the character start position of that word in <code>FNString</code> or returns <code>-1</code> if that word does not exist</li> <li>• For example: if <code>FNString</code> contains the string "hot and cold", <code>GetStart(3)</code> returns 9</li> </ul> |
| <code>GetWord()</code>  | <ul style="list-style-type: none"> <li>• Called with a parameter representing the position of the first character of a word in <code>FNString</code></li> <li>• Returns the word from <code>FNString</code></li> <li>• For example: if <code>FNString</code> contains the string "hot and cold", <code>GetWord(9)</code> returns "cold"</li> </ul>                                                                                       |

- (b) The programmer has decided to use a global ten-element 1D array `IgnoreList` of type `STRING` to store the ignored words. Unused elements contain the empty string ("") and may occur anywhere in the array.

A new module `AddWord()` is needed as follows:

| Module                 | Description                                                                                                                                                                                                                                                                                                                    |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>AddWord()</code> | <ul style="list-style-type: none"> <li>• Called with a parameter representing a word</li> <li>• Stores the word in an unused element of the <code>IgnoreList</code> array and returns <code>TRUE</code></li> <li>• Returns <code>FALSE</code> if the array was already full or if the word was already in the array</li> </ul> |

Write a detailed description of the algorithm for `AddWord()`. Do **not** include pseudocode statements in your answer.

**68. M/J 21/P22/Q1/b**

- 1 (a) (i) Complete the following table by giving the appropriate data type in each case.

| Variable | Example data value | Data type |
|----------|--------------------|-----------|
| Name     | "Catherine"        |           |
| Index    | 100                |           |
| Modified | FALSE              |           |
| Holiday  | 25/12/2020         |           |

[4]

- (ii) Evaluate each expression in the following table by using the initial data values shown in part (a)(i).

| Expression                 | Evaluates to |
|----------------------------|--------------|
| Modified OR Index > 100    |              |
| LENGTH("Student: " & Name) |              |
| INT(Index + 2.9)           |              |
| MID(Name, 1, 3)            |              |

[4]

- (b) Each pseudocode statement in the following table contains an example of selection, assignment or iteration.

Put **one** tick (✓) in the appropriate column for each statement.

| Statement               | Selection | Assignment | Iteration |
|-------------------------|-----------|------------|-----------|
| Index ← Index + 1       |           |            |           |
| IF Modified = TRUE THEN |           |            |           |
| ENDWHILE                |           |            |           |

[3]

**69. M/J 21/P22/Q2/b**

(b) A system is being developed to help manage book loans in a library.

Registered users may borrow books from the library for a period of time.

(i) State **three** items of data that must be stored for each loan.

1

2

3

[2]

(ii) State **one** item of data that will be required in the library system but does not need to be stored for each loan.

[1]

(iii) One operation that manipulates the data stored for each loan, would produce a list of all overdue books.

Identify **two other** operations.

Operation 1

Operation 2

[2]

**70. M/J 21/P22/Q4**

4 A teacher uses a paper-based system to store marks for a class test. The teacher requires a program to assign grades based on these results.

The program will output the grades together with the average mark.

Write a detailed description of the algorithm that will be needed.

[6]

**71. M/J 21/P23/Q1/a**

- 1 (a) A program is being developed to help manage the membership of a football club.

Complete the following identifier table.

| Example value | Explanation                                                                                         | Variable name | Data type |
|---------------|-----------------------------------------------------------------------------------------------------|---------------|-----------|
| "Wong"        | The preferred name of the member joining the football club                                          |               |           |
| FALSE         | A value to indicate whether an existing member of the club lives at the same address                |               |           |
| 19/02/1983    | When the member joined the football club                                                            |               |           |
| 1345          | The number of points a member has earned. Members of the club earn points for different activities. |               |           |

[4]

**72. M/J 21/P23/Q2/c**

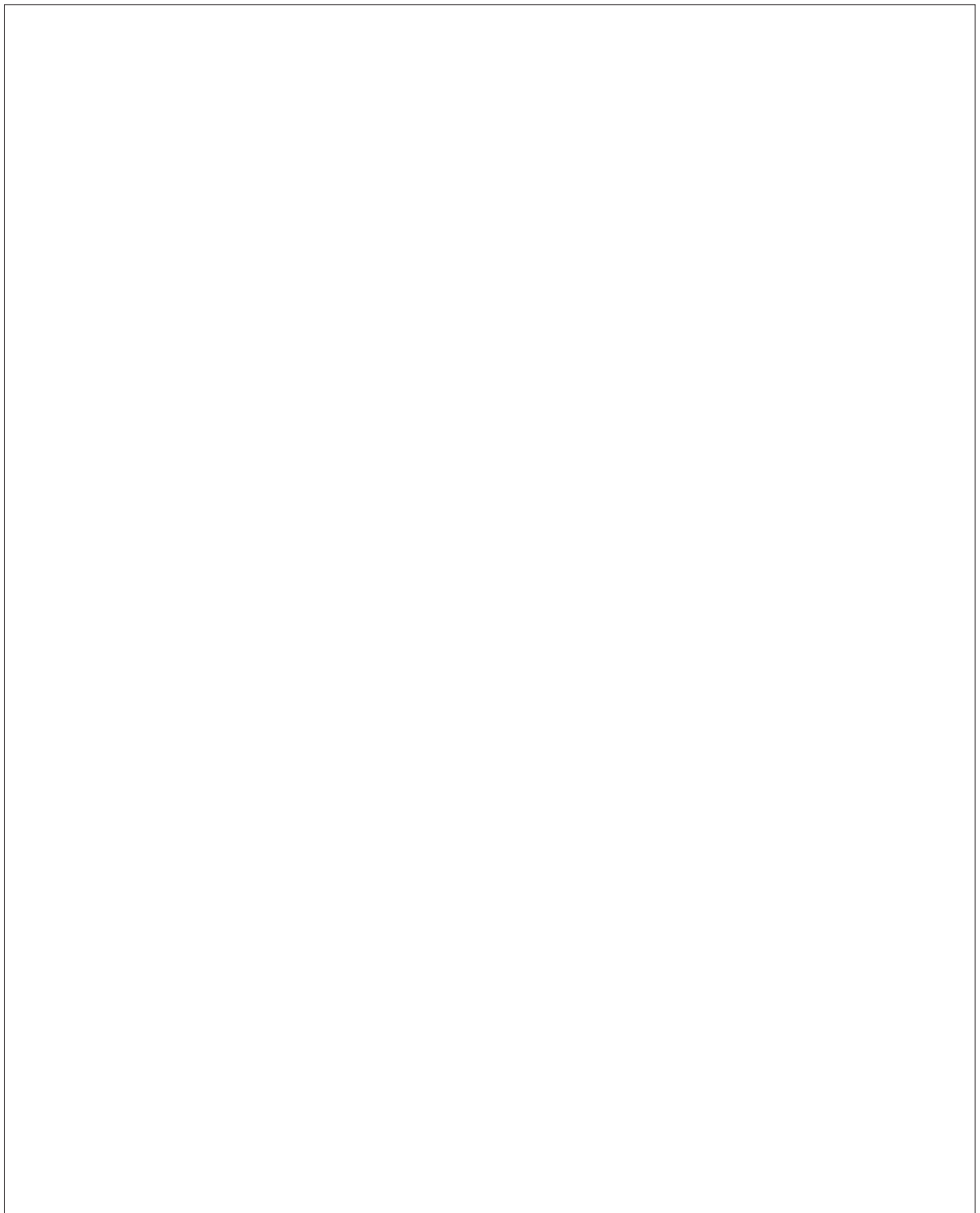
- (c) A program will:

- input 50 unique integer values
- output the largest value
- output the average of the values **excluding** the largest value.

Draw a program flowchart to represent the algorithm.

Variable declarations are **not** required.

It is not necessary to check that each input value is unique.



[6]

**73. M/J 21/P23/Q7/b**

**7** A program is needed to take a string containing a full name and produce a new string of initials.

Some words in the full name will be ignored. For example, "the", "and", "of", "for" and "to" may all be ignored.

Each letter of the abbreviated string must be upper case.

For example:

| Full name                                              | Initials |
|--------------------------------------------------------|----------|
| Integrated Development Environment                     | IDE      |
| The American Standard Code for Information Interchange | ASCII    |

The programmer has decided to use a global variable `FNString` of type `STRING` to store the full name.

It is assumed that:

- words in the full name string are separated by a single space character
- space characters will not occur at the beginning or the end of the full name string
- the full name string contains at least one word.

The programmer has started to define program modules as follows:

| Module                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>GetStart()</code> | <ul style="list-style-type: none"> <li>• Called with an <code>INTEGER</code> as a parameter, representing the number of a word in <code>FNString</code>.</li> <li>• Returns the character start position of that word in <code>FNString</code> or returns <code>-1</code> if that word does not exist</li> <li>• For example: if <code>FNString</code> contains the string "hot and cold", <code>GetStart(3)</code> returns 9</li> </ul> |
| <code>GetWord()</code>  | <ul style="list-style-type: none"> <li>• Called with a parameter representing the position of the first character of a word in <code>FNString</code></li> <li>• Returns the word from <code>FNString</code></li> <li>• For example: if <code>FNString</code> contains the string "hot and cold", <code>GetWord(9)</code> returns "cold"</li> </ul>                                                                                       |

- (b) The programmer has decided to use a global ten-element 1D array `IgnoreList` of type `STRING` to store the ignored words. Unused elements contain the empty string ("") and may occur anywhere in the array.

A new module `AddWord()` is needed as follows:

| Module                 | Description                                                                                                                                                                                                                                                                                                                    |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>AddWord()</code> | <ul style="list-style-type: none"> <li>• Called with a parameter representing a word</li> <li>• Stores the word in an unused element of the <code>IgnoreList</code> array and returns <code>TRUE</code></li> <li>• Returns <code>FALSE</code> if the array was already full or if the word was already in the array</li> </ul> |

Write a detailed description of the algorithm for `AddWord()`. Do **not** include pseudocode statements in your answer.

## Answers Section

### 1. M/J 26/P21/Q2/a(ii,iii)

|           |                                                                                                                                                                                                                                          |   |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 2(a)(ii)  | <p>MainsOn (= TRUE) AND WaterOn (= TRUE)</p> <p>Or reverse order ...</p>                                                                                                                                                                 | 1 |
| 2(a)(iii) | <p>IF<br/>MainsOn AND (Tray1 OR Tray2) THEN</p> <div style="text-align: center;"> </div> <p>WashStart ← TRUE</p> <p>ENDIF</p> <p><b>OR:</b></p> <div style="text-align: center;"> </div> <p>WashStart ← MainsOn AND (Tray1 OR Tray2)</p> | 3 |

### 2. M/J 26/P21/Q4

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 4 | <p><i>Example solution:</i></p> <p>Initialise Count to zero<br/>REPEAT<br/>  INPUT integer number<br/>  IF integer number &gt; 0 then increment the <b>count</b><br/>  UNTIL integer number equals 0<br/>OUTPUT Count</p> <p><i>Mark as follows:</i></p> <p><b>MP1</b>    Initialise Count to zero <b>and</b> later evidence of use of counter<br/> <b>MP2</b>    Conditional loop <b>and</b> terminated when 0 is input<br/> <b>MP3</b>    Test for number &gt; 0 <b>and</b> increment count <b>inside the loop</b><br/> <b>MP4</b>    Output the count (either 'sum' or 'count')</p> <p><b>Max 4</b></p> | 4 |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

**3. M/J 26/P22/Q2/a,b,c**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |                                                                                                          |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------|
| 2(a) | <p>One mark for either explanation:</p> <ul style="list-style-type: none"> <li>To filter out <u>information / data</u> that is <b>not</b> necessary for the task</li> <li>To keep only <u>information / data</u> that <b>is</b> necessary for the task</li> </ul> <p><b>Max 1 mark</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>1</b> |                                                                                                          |
| 2(b) | <p>One mark for each <b>TWO</b> data items</p> <p>Example answers</p> <ol style="list-style-type: none"> <li>1 Tool ID // Tool name // Type of tool</li> <li>2 Customer ID // Customer name // Customer address // Account reference</li> <li>3 Hire ID</li> <li>4 (Hire) Date</li> <li>5 Hire time</li> <li>6 Length of hire: Full or Half day</li> <li>7 Cost of hire</li> <li>8 Payment details</li> <li>9 Tool status</li> </ol> <p><b>Max 2 marks</b></p>                                                                                                                                                                                                                                                           | <b>2</b> | <p>Reject Quantity<br/>Reject Return date if Hire date already given<br/>MP 2 Reject just 'Customer'</p> |
| 2(c) | <p>One mark for each process.</p> <p>Examples include:</p> <ol style="list-style-type: none"> <li>1 Input customer details / customer ID / payment details / Tool ID</li> <li>2 Search for required tool // Checking if tool returned // Checking if tool available // Searching for dates when tool is available</li> <li>3 Record tool booking // Amend Tool booking // Create tool booking // Record when tool returned // Confirm booking</li> <li>4 Calculating <b>time</b> when tool has to be returned</li> <li>5 Change / check tool <b>status</b> e.g. hired / available / damaged // Prevent same tool being hired twice</li> <li>6 Process payment / calculate hire cost</li> </ol> <p><b>Max 2 marks</b></p> | <b>2</b> |                                                                                                          |

**4. M/J 26/P22/Q3**

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                          |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 | <p>Example solution</p> <p>Mark points:</p> <p>MP 1 (Step 4) Open <u>WordList.txt</u> / File in <b>read</b> mode (and subsequently close)</p> <p>MP 2 (Step 5) <b>Read</b> the next <b>line</b> from the <b>file</b> in a loop</p> <p>MP 3 (Step 6) Increment line counter in a loop</p> <p>MP 4 (Step 7) Check if the line read equals the word (input) in a loop</p> <p>MP 5 (Step 8) ...If it does output line counter</p> <p>MP 6 (Step 9) If word not found in file then output "Not Found" after all iterations</p> <p>MP 7 (Step 10) Otherwise repeat from step 5</p> <p><b>Max 6 marks from 7 possible marks</b></p> | <p><b>6</b></p> <p>MP 1 cannot be in a loop structure</p> <p>Do not accept once if MP 2, 3, 4 is <b>clearly</b> not in a loop</p> <p>Award only one mark for MP 5 and MP 6 if <b>both</b> outputs are possible</p> <p>MP 7 Can be any loop but it <b>must be clear</b> where it starts and ends and must include at least one of the end conditions.</p> |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**5. M/J 26/P23/Q2/b,c**

|      |                                                                                                                                                                                  |   |                                                                  |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------------------------------------------------------------------|
| 2(b) | <pre>IF MainsOn (=TRUE) THEN     DisplayOn ← TRUE ENDIF Accept: DisplayOn ← MainsOn</pre> <p>Accept '=' in place of the assignment arrow</p>                                     | 1 |                                                                  |
| 2(c) | <p><b>MP1</b> The robot is only ready to move if... the <b>mains power is on and .....</b></p> <p><b>MP2</b> The strength of the <b>(suction) value selected is non-zero</b></p> | 2 | <p>Reject – Zero for a 'repeat' of the pseudocode statement.</p> |

**6. M/J 26/P23/Q4**

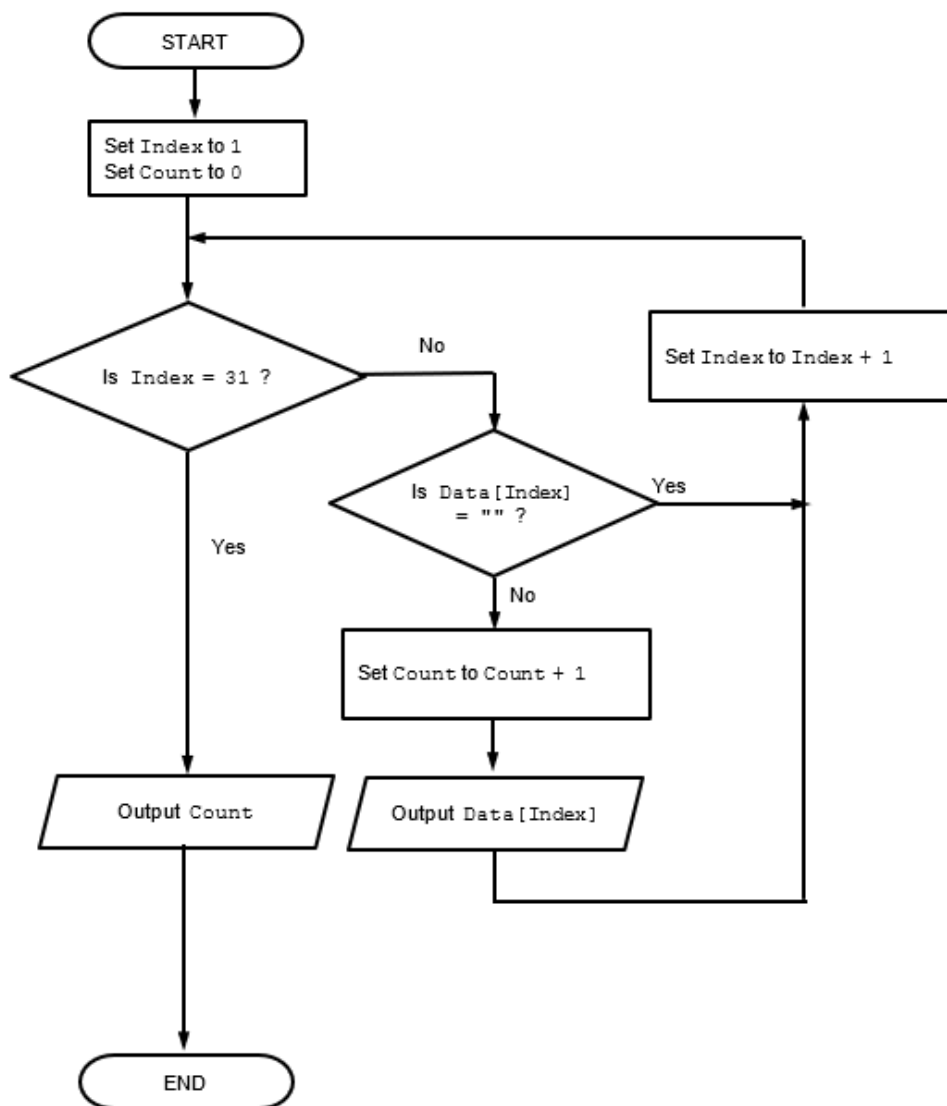
|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |                                            |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--------------------------------------------|
| 4 | <p><i>Example solution:</i></p> <ul style="list-style-type: none"> <li>• Initialise count to zero</li> <li>• REPEAT</li> <li>• Input next number</li> <li>• If number &gt; 120 then increment the count</li> <li>• Until number = -1</li> <li>• Output the count</li> </ul> <p><i>Mark as follows:</i></p> <p><b>MP1</b> Initialise count <b>and</b> later evidence of use of counter (<b>not</b> 'sum')</p> <p><b>MP2</b> Conditional loop and terminated when -1 is input</p> <p><b>MP3</b> Test for number &gt; 120 <b>and</b> increment the count (<b>not</b> 'sum') <b>inside the loop</b></p> <p><b>MP4</b> Output the count (Accept either 'sum' or 'count')</p> | 4 | <p>A 'sum' solution will have a Max 2.</p> |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--------------------------------------------|

## 7. O/N 25/P21/Q2

2

Example solution

5



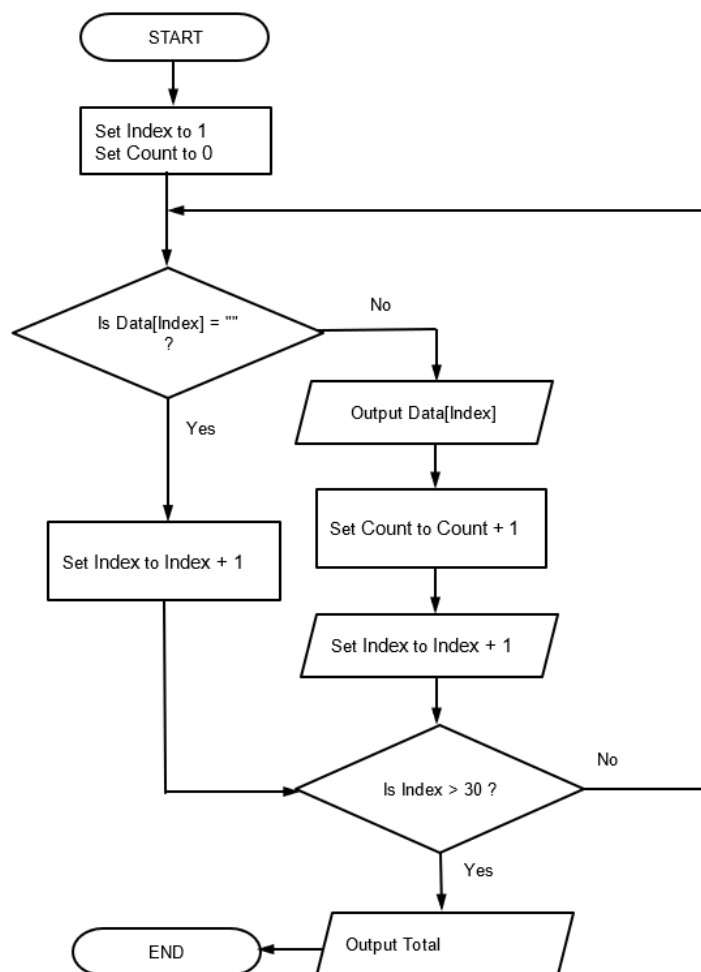
Mark as follows:

- 1 Initialisation of variables used as **array index** and **count**
- 2 Loop through exactly 30 elements in the array
- 3 Test for empty string in current array element // Test for non-empty string
- 4 Increment count if appropriate **in a loop**
- 5 Both required outputs under correct conditions with correct outputs

2

## Alternative solution

Checking for empty string first



## 8. O/N 25/P22/Q1/a

1(a)

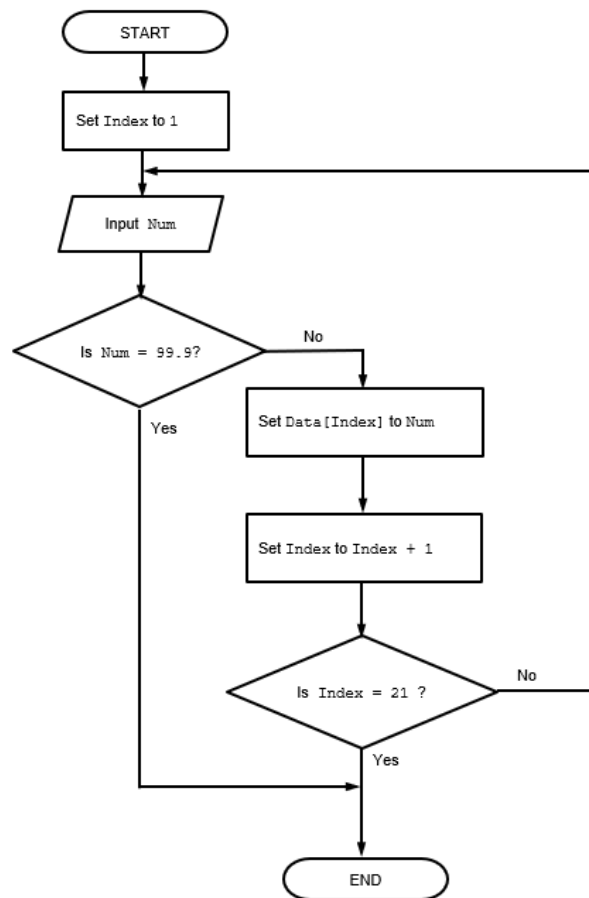
| Pseudocode example                                                                         | Selection | Iteration | Subroutine |
|--------------------------------------------------------------------------------------------|-----------|-----------|------------|
| IF Status = FALSE THEN<br>FOR Count ← 1 TO 20<br>CALL Reset (Count)<br>NEXT Count<br>ENDIF | ✓         | ✓         | ✓          |
| OTHERWISE : Status ← TRUE                                                                  | ✓         |           |            |
| WHILE AllDone() = TRUE                                                                     |           | ✓         | ✓          |
| 30 : NextChar ← 'X'                                                                        | ✓         |           |            |

One mark per row

4

## 9. O/N 25/P22/Q2

2



5

Mark as follows:

**MP1** Initialisation of the array index

**MP2** Input number **in a loop**

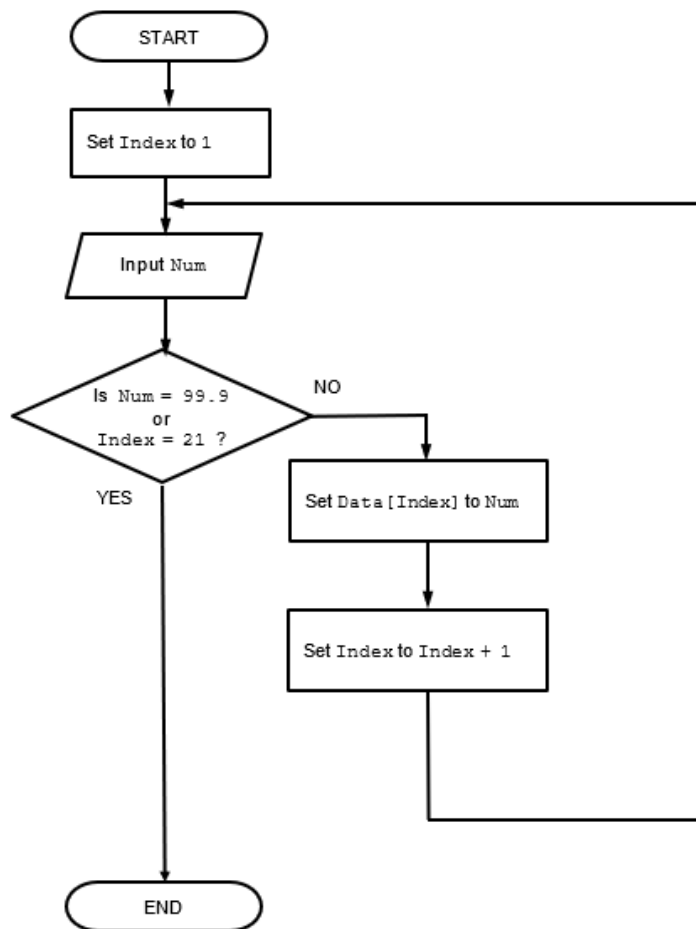
**MP3** Test for termination value (99.9) **and** correct ending

**MP4** Test for array full / maximum 20 iterations **and** correct ending

**MP5** Assign number to Data array element **and** increment index

2

Alternative solution:



## 10. O/N 25/P22/Q3/a

3(a)

Mark as follows:

6

|            |   |  |                                                                                       |
|------------|---|--|---------------------------------------------------------------------------------------|
| <b>MP1</b> | 1 |  | open the file <b>OldFile.txt</b> in <b>read (mode)</b>                                |
| <b>MP2</b> | 2 |  | open the file <b>NewFile.txt</b> in <b>write (mode)</b>                               |
|            | 3 |  | read a line from <b>OldFile.txt</b> and store in <b>LineX</b>                         |
| <b>MP3</b> | 4 |  | read a line from <b>OldFile.txt</b> and store in <b>LineY</b>                         |
|            | 5 |  | read a line from <b>OldFile.txt</b> and store in <b>LineZ</b>                         |
| <b>MP4</b> | 6 |  | set <b>NewString</b> to<br><b>LineX &amp; '\ ' &amp; LineY &amp; '\ ' &amp; LineZ</b> |
| <b>MP5</b> | 7 |  | write <b>NewString</b> to <b>NewFile.txt</b>                                          |
| <b>MP6</b> | 8 |  | repeat from step <b>3</b> until <b>end of OldFile.txt</b>                             |
|            | 9 |  | close both files                                                                      |

**11. O/N 25/P22/Q4/c**

|      |                                                                                                                                                                                                                                                                         |          |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 4(c) | <b>MP1</b> Check that the parameter value is in the range 0 to 35<br><b>MP2</b> ... and if not, return FALSE<br><b>MP3</b> Use the <b>parameter</b> value as <b>the index to array</b> <code>CheckTotal</code><br><b>MP4</b> Return the value from given index position | <b>4</b> |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

**12. O/N 25/P23/Q2/a,b**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2(a) | <p>One mark per point:</p> <ol style="list-style-type: none"> <li>1 Open the file in append mode (and subsequently close) <b>before loop</b></li> <li>2 Initialise a variable to 1 / 0 <b>before loop</b></li> <li>3 Use the variable as an index to the array to access an element <b>in a loop</b></li> <li>4 Test if element value is blank / empty <b>in a loop</b></li> <li>5 If not blank then write the value to the file <b>in a loop</b></li> <li>6 Increment the variable <b>in a loop</b></li> <li>7 Repeat from MP3 until variable value is 66 / 65 / all elements (in the array) have been checked</li> </ol> <p><b>Max 6</b></p> | <b>6</b> |
| 2(b) | <p>One mark per point:</p> <p>Construct: Selection</p> <p>Use: To test whether the (array) element is blank</p> <p>Alternative:</p> <p>Construct: Sequence</p> <p>Use: To specify the order in which the steps of the algorithm have to be followed so that the task is completed correctly</p>                                                                                                                                                                                                                                                                                                                                                | <b>2</b> |

**13. M/J 25/P21/Q1/a,b,c**

1(a)

| Example value | Explanation                                  | Variable name     | Data type |
|---------------|----------------------------------------------|-------------------|-----------|
| "Fruit"       | a category of stock that is sold in the shop | Category          | STRING    |
| 20/02/2025    | when an item was sold                        | DateSold          | DATE      |
| 12.67         | the cost of an item                          | (Item) Cost/Price | REAL      |
| TRUE          | to indicate if an item is in stock           | (Item) InStock    | BOOLEAN   |

Mark as follows:  
1 mark per row for each variable name **and** data type

4

|      |                                                           |                   |                  |                  |   |
|------|-----------------------------------------------------------|-------------------|------------------|------------------|---|
| 1(b) | <b>Pseudocode extract</b>                                 | <b>Assignment</b> | <b>Selection</b> | <b>Iteration</b> | 5 |
|      | Result ← CalculateTotal()                                 | ✓                 |                  |                  |   |
|      | WHILE IsClosed                                            |                   |                  | ✓                |   |
|      | REPEAT<br>INPUT Value<br>UNTIL Sales[4] > Value           | ✓                 |                  | ✓                |   |
|      | IF Sales[Current] <= 150 THEN<br>Discount ← TRUE<br>ENDIF | ✓                 | ✓                |                  |   |
|      | CASE OF Option                                            |                   | ✓                |                  |   |
|      | Mark as follows:<br>One mark for each correct row         |                   |                  |                  |   |

|                                             |                         |                                                                                                                                                                            |  |  |   |
|---------------------------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|---|
| 1(c)                                        | Decomposition involves: |                                                                                                                                                                            |  |  | 3 |
|                                             | <b>MP1</b>              | Breaking down a problem into <b>sub problems</b>                                                                                                                           |  |  |   |
|                                             | <b>MP2</b>              | In order to explain / understand <b>the process/task</b> of how a (stock control) can be managed in more detail                                                            |  |  |   |
|                                             | <b>MP3</b>              | Leading to the concept of the program designed as a series of <b>modules</b> / procedures / functions<br>// or <b>by example</b> of different <b>stock control</b> modules |  |  |   |
|                                             | <b>MP4</b>              | Makes the <b>program</b> easier to <b>test / debug</b>                                                                                                                     |  |  |   |
|                                             | <b>Max 3</b>            |                                                                                                                                                                            |  |  |   |
| <b>Max 2</b> if no mention of stock control |                         |                                                                                                                                                                            |  |  |   |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 3 | <p><b>Example solution:</b></p> <pre> DECLARE GeneratedValue : INTEGER DECLARE Guess : INTEGER  GeneratedValue ← INT(RAND(100)) + 1  REPEAT   OUTPUT "Enter a guess between 1 and 100: "   INPUT Guess    IF Guess &gt; GeneratedValue THEN     OUTPUT "Guess was too high"   ENDIF   IF Guess &lt; GeneratedValue THEN     OUTPUT "Guess was too low"   ENDIF UNTIL Guess = GeneratedValue  OUTPUT "Number guessed correctly"  <b>MP1</b>   Declare all variables used  <b>MP2</b>   Uses RAND() function <b>MP3</b>   Uses INT() function <b>MP4</b>   Correct syntax for random number between 1 and 100  <b>MP5</b>   Conditional loop until correct number guessed <b>MP6</b>   Prompt and input a guess <b>in a loop</b>  <b>MP7</b>   Check if guess is too high <b>in a loop</b> <b>MP8</b>   Check if guess is too low <b>in a loop</b>  <b>MP9</b>   Output appropriate message (x2) when the number is <b>not</b> guessed correctly         (in the loop) <b>and</b> output correct guess message  <b>Max 8</b> </pre> | 8 |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

**15. M/J 25/P22/Q2/a**

|      |                                                                                                                                                                                                                                                                                                         |   |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 2(a) | <p>Mark as follows:</p> <ol style="list-style-type: none"> <li>1. To increase the level of detail of the algorithm // To break the problem / task into <b>smaller steps</b> ...</li> <li>2. ... until steps can be directly translated into lines of code // from which it can be programmed</li> </ol> | 2 |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

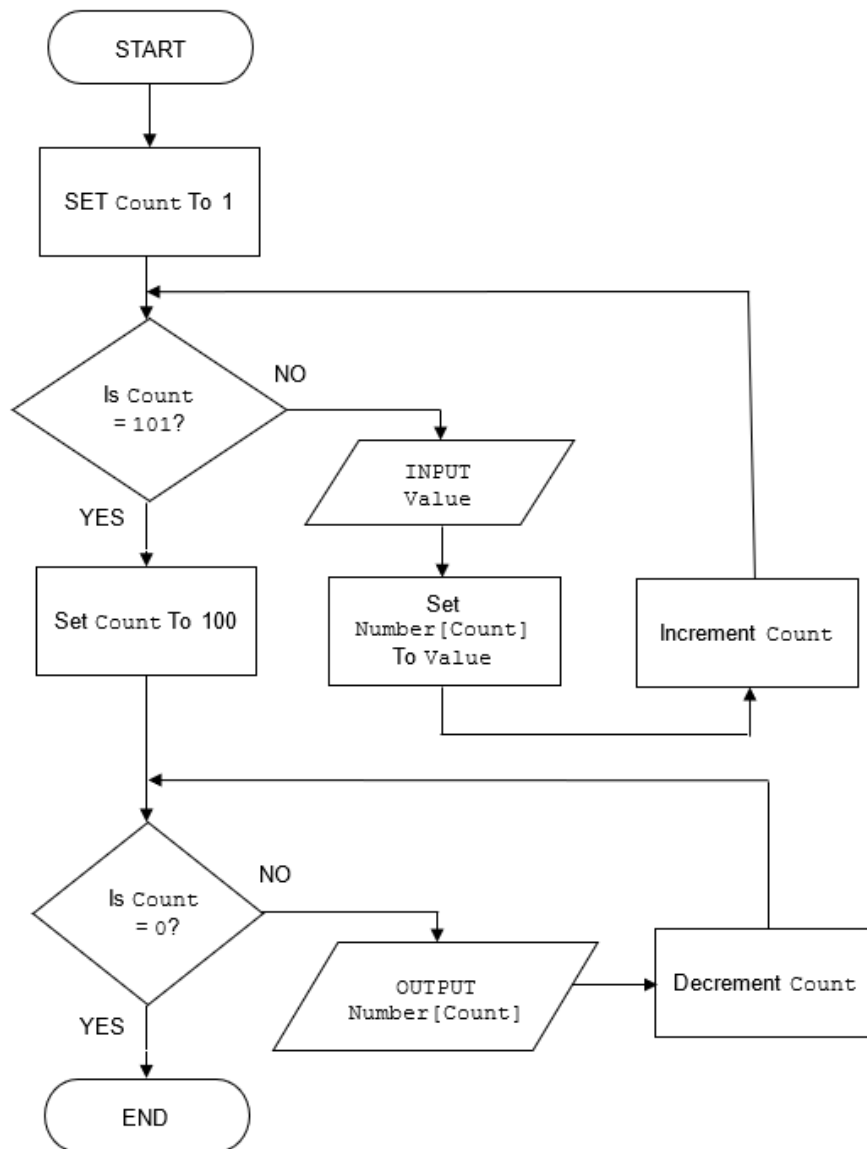
**16. M/J 25/P22/Q3**

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 3 | <p><b>Example solution</b></p> <pre> DECLARE FirstNumber : INTEGER DECLARE SecondNumber : INTEGER DECLARE ThirdNumber : INTEGER  FirstNumber ← INT(RAND(21)) - 10  OUTPUT FirstNumber  REPEAT     SecondNumber ← INT(RAND(21)) - 10 UNTIL FirstNumber &lt;&gt; SecondNumber  OUTPUT SecondNumber  IF FirstNumber &lt; 0 AND SecondNumber &lt; 0 THEN     ThirdNumber ← INT(RAND(6)) + 30     OUTPUT ThirdNumber ENDIF </pre> <p><b>Mark points:</b></p> <ol style="list-style-type: none"> <li>1. Declare all variables used</li> <li>2. Use <code>RAND()</code> function with any <b>integer</b> parameter</li> <li>3. Use <code>INT()</code> function with any <b>numeric</b> parameter</li> <li>4. Conditional loop until <b>two different</b> random numbers are generated</li> <li>5. Use <code>INT()</code> function using random number generated between –10 and 10 inclusive <b>in a loop</b></li> <li>6. Output <b>two different</b> random integers / numbers</li> <li>7. Check if both random integers / numbers are negative</li> <li>8. then output a (third) random integer / number between 30 and 35 inclusive</li> </ol> | 8 |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

**17. M/J 25/P22/Q4**

4

6



One mark for each functional group as listed:

1. Initialise Index used for `Number` to either 1 or 0 before first loop
2. Loop 100 times
3. Input value
4. Increment array index **and** store value in `Number` array at element referenced by array index **in a loop**
5. Output loop starts at last element in `Number` array
6. Output `Number` array element referenced by Index **in a loop**
7. Output all elements of `Number` array **once** in reverse order

**Max 6**

**18. M/J 25/P23/Q3**

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 3 | <p><b>Example solution:</b></p> <pre> FUNCTION RollDice (NoOfTimes : INTEGER) RETURNS REAL   DECLARE RollValue : INTEGER   DECLARE Average : REAL   DECLARE Count : INTEGER   DECLARE Sum : INTEGER    Sum ← 0    FOR Count ← 1 TO NoOfTimes     RollValue ← INT(RAND(6)) + 1     OUTPUT RollValue     Sum ← Sum + RollValue   NEXT Count    Average ← Sum / NoOfTimes   RETURN Average ENDFUNCTION </pre> <p><b>MP1</b> Declare all variables used <b>and</b> initialise Sum to 0</p> <p><b>MP2</b> Loop for 'number of times' parameter</p> <p><b>MP3</b> Use RAND() function with Integer parameter <b>in a loop</b></p> <p><b>MP4</b> Use INT() function <b>in a loop</b></p> <p><b>MP5</b> Generate a random integer between 1 and 6 <b>in a loop</b></p> <p><b>MP6</b> Output each value generated <b>in a loop</b></p> <p><b>MP7</b> Calculate Sum <b>and</b> if Average used check declared as REAL <b>and</b> return Average <b>and</b> check function header returns REAL</p> | 7 |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

**19. M/J 25/P23/Q4/a**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 4(a) | <p><b>MP1</b> Open the file (Sales.txt) in <b>read mode</b> and subsequently close</p> <p><b>MP2</b> <b>Read a line</b> (from the text file)</p> <p><b>MP3</b> Convert the line read (string) to a number</p> <p><b>MP4</b> If the number is greater than 500 <b>and</b> then output week number</p> <p><b>MP5</b> Increment week number (must have been Initialised ...)</p> <p><b>MP6</b> Repeat from step 2 for 52 times // Repeat from step 2 until end of file</p> <p><b>Max 5</b></p> | 5 |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

**20. O/N 24/P21/Q3/a,b**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 3(a) | <p>One mark per emboldened part:</p> <ol style="list-style-type: none"> <li>1 Open the file <b>(Result.txt)</b> in <u>read</u> mode</li> <li>2 Loop until <b>EOF(Result.txt) // EOF // end of file</b></li> <li>3 <b>Read a / the / next line (from the file)</b> and store in ThisLine</li> <li>4 Assign LineY to LineX</li> <li>5 Assign LineZ to LineY</li> <li>6 Assign ThisLine to LineZ</li> <li>7 After the loop, <b>close the file (Result.txt)</b></li> <li>8 Output LineX, LineY, LineZ</li> </ol> | <b>4</b> |
| 3(b) | <p>Example answer:</p> <p>So that the last three lines of the file are output in the correct <b>order</b></p> <p>A mark for mentioning one of:</p> <ul style="list-style-type: none"> <li>• (Ensuring) the lines are <b>output in correct order</b></li> <li>• Ordering the last <u>three</u> lines</li> <li>• Ordering the lines so they are in the same order as (they occur) <u>in the file</u></li> </ul> <p><b>Max 1 marks</b></p>                                                                      | <b>1</b> |

**21. O/N 24/P22/Q7/a,b**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 7(a) | <ul style="list-style-type: none"> <li>• Customer ID – to reference the other customer details</li> <li>• Email address – to send the email</li> <li>• Name – to personalise the email</li> <li>• Date of last visit – to select which customers should receive an email</li> <li>• Unique voucher code (or method of code generation) – to include in the email</li> </ul> <p>Mark as follows:</p> <p>One mark per item <b>and</b> justification</p> <p><b>Max 3 marks</b></p> | <b>3</b> |
| 7(b) | Abstraction                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>1</b> |

**22. O/N 24/P23/Q1/a**

|                  |                                                                                            |           |           |            |   |
|------------------|--------------------------------------------------------------------------------------------|-----------|-----------|------------|---|
| 1(a)             | Pseudocode example                                                                         | Selection | Iteration | Subroutine | 4 |
|                  | FOR Index ← 1 TO 3<br>IF Safe[Index] = TRUE THEN<br>Flap[Index] ← 0<br>ENDIF<br>NEXT Index | ✓         | ✓         |            |   |
|                  | CASE OF Compound(3)                                                                        | ✓         |           | ✓          |   |
|                  | REPEAT UNTIL AllDone() = TRUE                                                              |           | ✓         | ✓          |   |
|                  | WHILE Result[3] <> FALSE                                                                   |           | ✓         |            |   |
| One mark per row |                                                                                            |           |           |            |   |

**23. O/N 24/P23/Q2/a,b**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 2(a) | <p>DECLARE Count, Total, NextNumber : INTEGER</p> <p>Total ← 0</p> <p>FOR Count ← 1 TO 100</p> <p>  OUTPUT "Input an integer value"</p> <p>  INPUT NextNumber</p> <p>  IF NextNumber &gt; 0 THEN</p> <p>    Total ← Total + NextNumber</p> <p>  ENDIF</p> <p>NEXT Count</p> <p>OUTPUT Total</p> <p>Mark as follows:</p> <p><b>MP1</b> Declarations of all variables used</p> <p><b>MP2</b> Loop for 100 iterations</p> <p><b>MP3</b> Prompt <b>and</b> input a value <b>in a loop and</b></p> <p><b>MP4</b> Test for value &gt; 0 // &gt;=1 <b>in a loop</b></p> <p><b>MP5</b> Sum the Total <b>in a loop</b></p> <p><b>MP6</b> Output of Total after the <b>loop</b></p> <p><b>Max 5 marks</b></p> | 5 |
| 2(b) | <p><b>MP1</b> Construct: Iteration / Repetition</p> <p><b>MP2</b> Use: To loop through all <u>100</u> inputs // To loop <u>100</u> times</p> <p>ALTERNATIVE:</p> <p><b>MP1</b> Construct: Selection</p> <p><b>MP2</b> Use: To test whether the value input is positive</p>                                                                                                                                                                                                                                                                                                                                                                                                                          | 2 |

**24. O/N 24/P23/Q3/b,c**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 3(b) | <pre>graph TD; START([START]) --&gt; SET1[SET Index TO 1]; SET1 --&gt; IS200{IS Index = 200?}; IS200 -- NO --&gt; SETPTR[SET List[Index].Pointer TO Index + 1]; SETPTR --&gt; SETINC[SET Index TO Index + 1]; SETINC --&gt; IS200; IS200 -- YES --&gt; SET0[SET List[Index].Pointer TO 0]; SET0 --&gt; END([END]);</pre> <p>Mark as follows:</p> <ul style="list-style-type: none"><li><b>MP1</b> Use of variable as array index <b>and</b> initialisation to 1 / to first element of array</li><li><b>MP2</b> Loop for 199 / 200 iterations</li><li><b>MP3</b> Assign Index+1 to each pointer field in a loop</li><li><b>MP4</b> Assign 0 to 200<sup>th</sup> pointer field</li></ul> | <b>4</b> |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 3(c) | <p>One mark per step:</p> <p><b>MP1</b> Assign <code>HeadPointer</code> to <code>ThisPointer</code> / Current Pointer // Identify first node using <u>headpointer</u></p> <p><b>MP2</b> Loop the following until <code>ThisPointer</code> value is 0 (zero) / null <u>pointer</u> reached</p> <p><b>MP3</b> ... Output data (field) from array element at index <code>ThisPointer</code> // Output data (field) of the current node / array element</p> <p><b>MP4</b> ... Assign pointer field from current array element / index / to <code>ThisPointer</code> / Current Pointer // Assign pointer field from current node to <code>ThisPointer</code> / Current Pointer</p> | <b>4</b> |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

**25. O/N 24/P23/Q7/b**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 7(b) | <p>One mark for name <b>and</b> use</p> <p>Mark as follows:<br/>Examples include:</p> <p>Sub-module: <code>ScanCard()</code> / <code>GetID()</code><br/>Use: Read the barcode from the loyalty card / Get the customer ID from the barcode</p> <p>Sub-module: <code>GetPoints()</code><br/>Use: Get the number of points the customer has</p> <p>Sub-module: <code>ExchangePoints()</code> / <code>UpdateCard()</code><br/>Use: Reduce the number of loyalty points</p> <p>Sub-module: <code>GetDiscount()</code> / <code>CalculateDiscount()</code><br/>Use: Calculate the discount</p> <p>Sub-module: <code>CalculatePoints()</code><br/>Use: Calculate points (following a purchase)</p> <p>Sub-module: <code>UpdatePoints()</code><br/>Use: Update total points a customer has (following a purchase)</p> <p>Sub-module: <code>ShowDiscount()</code><br/>Use: Display the discount</p> <p><b>Max 4 marks</b></p> | <b>4</b> |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

**26. M/J 24/P21/Q1/c**

|          |                                                                                                 |          |
|----------|-------------------------------------------------------------------------------------------------|----------|
| 1(c)(i)  | The names do not reflect / indicate the purpose of the variable // the names are not meaningful | <b>1</b> |
| 1(c)(ii) | They make the program more difficult to understand / debug / maintain                           | <b>1</b> |

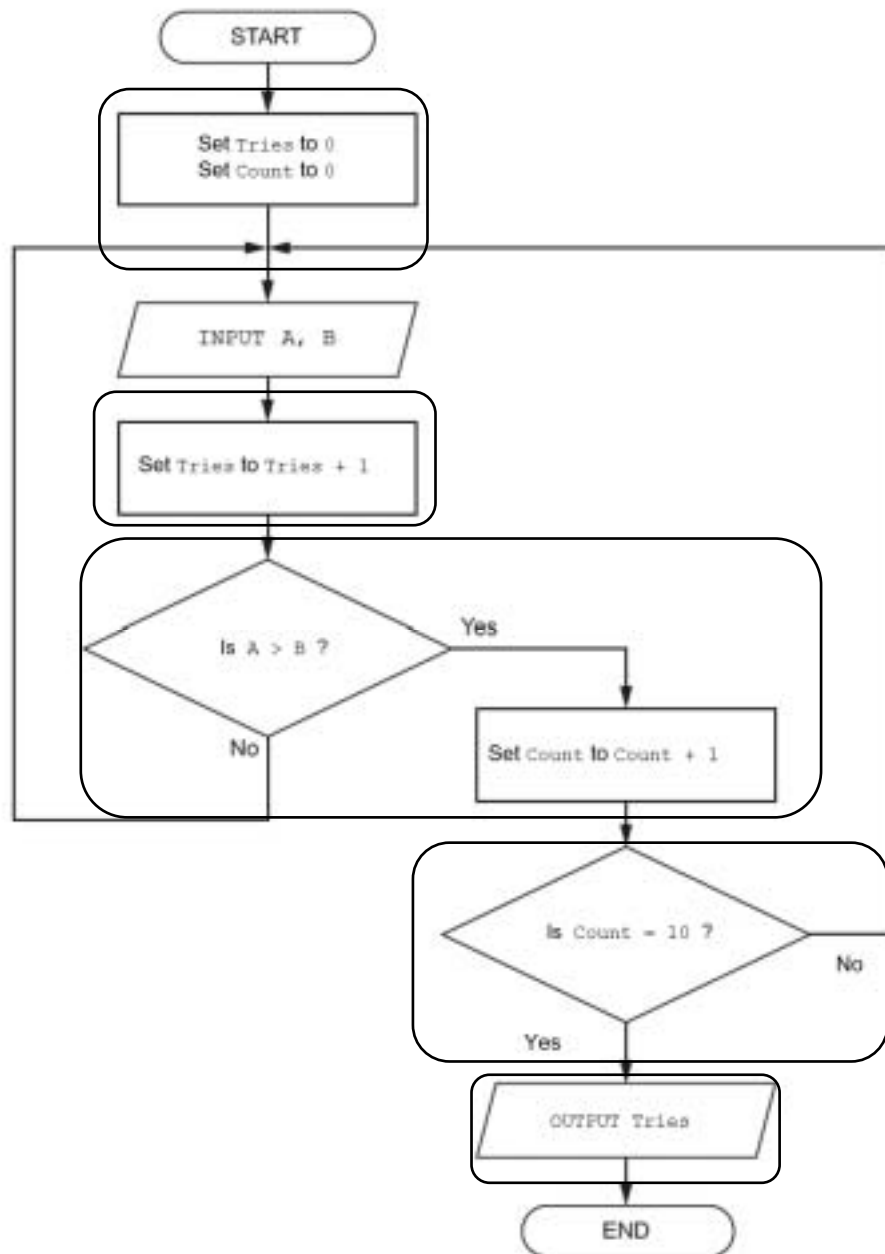
|           |                                                                                                                                                                                                                                                                                    |          |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1(c)(iii) | <p>One mark from:</p> <ul style="list-style-type: none"> <li>• Indentation / use of white space</li> <li>• Capitalisation of keywords</li> <li>• Use of comments</li> <li>• Use of modular programming</li> <li>• Use of local variables</li> </ul> <p><b>Note: max 1 mark</b></p> | <b>1</b> |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

| 1(a)             | <table><tr><th>Assignment statement</th><th>Data type</th></tr><tr><td>A ← LEFT(MyName, 1)</td><td>CHAR / STRING</td></tr><tr><td>B ← Total * 2</td><td>INTEGER / REAL</td></tr><tr><td>C ← INT(ItemCost) / 3</td><td>REAL</td></tr><tr><td>D ← "Odd OR Even"</td><td>STRING</td></tr></table> | Assignment statement | Data type | A ← LEFT(MyName, 1) | CHAR / STRING | B ← Total * 2 | INTEGER / REAL | C ← INT(ItemCost) / 3 | REAL | D ← "Odd OR Even" | STRING | 4 |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------|---------------------|---------------|---------------|----------------|-----------------------|------|-------------------|--------|---|
|                  | Assignment statement                                                                                                                                                                                                                                                                           | Data type            |           |                     |               |               |                |                       |      |                   |        |   |
|                  | A ← LEFT(MyName, 1)                                                                                                                                                                                                                                                                            | CHAR / STRING        |           |                     |               |               |                |                       |      |                   |        |   |
|                  | B ← Total * 2                                                                                                                                                                                                                                                                                  | INTEGER / REAL       |           |                     |               |               |                |                       |      |                   |        |   |
|                  | C ← INT(ItemCost) / 3                                                                                                                                                                                                                                                                          | REAL                 |           |                     |               |               |                |                       |      |                   |        |   |
|                  | D ← "Odd OR Even"                                                                                                                                                                                                                                                                              | STRING               |           |                     |               |               |                |                       |      |                   |        |   |
| One mark per row |                                                                                                                                                                                                                                                                                                |                      |           |                     |               |               |                |                       |      |                   |        |   |

**27. M/J 24/P21/Q2/a**

2(a)

5



One mark per outlined region:

- 1 Initialise both counts
- 2 Increment `Tries` every time a pair is input
- 3 Compare `A > B` **and** increment `Count` if TRUE
- 4 Test for `Count = 10` (10<sup>th</sup> time `A > B`) – MUST include Yes / No labels
- 5 If so output `Tries`, otherwise loop

## 28. M/J 24/P21/Q7/a

|         |                                                           |   |
|---------|-----------------------------------------------------------|---|
| 7(a)(i) | To make the solution easier to design / implement / solve | 1 |
|---------|-----------------------------------------------------------|---|

|           |                                                                                                                                                                                                                                                                                                                 |          |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 7(a)(ii)  | <p>One mark per item <b>and</b> justification</p> <p>Item: mobile phone number<br/>Justification: to send the text message</p> <p>Item: name<br/>Justification: to personalise the text message</p> <p>Item: exercise interest<br/>Justification: to determine whether this member would be interested</p>      | <b>3</b> |
| 7(a)(iii) | <p>Examples include:</p> <ul style="list-style-type: none"> <li>• Add a member to a list of those interested in the new class</li> <li>• Remove the member from future SMS messages</li> <li>• Read/process Message</li> <li>• Identify who from</li> </ul> <p>One mark for each.</p> <p><b>Max 2 marks</b></p> | <b>2</b> |

**29. M/J 24/P22/Q1/a,c**

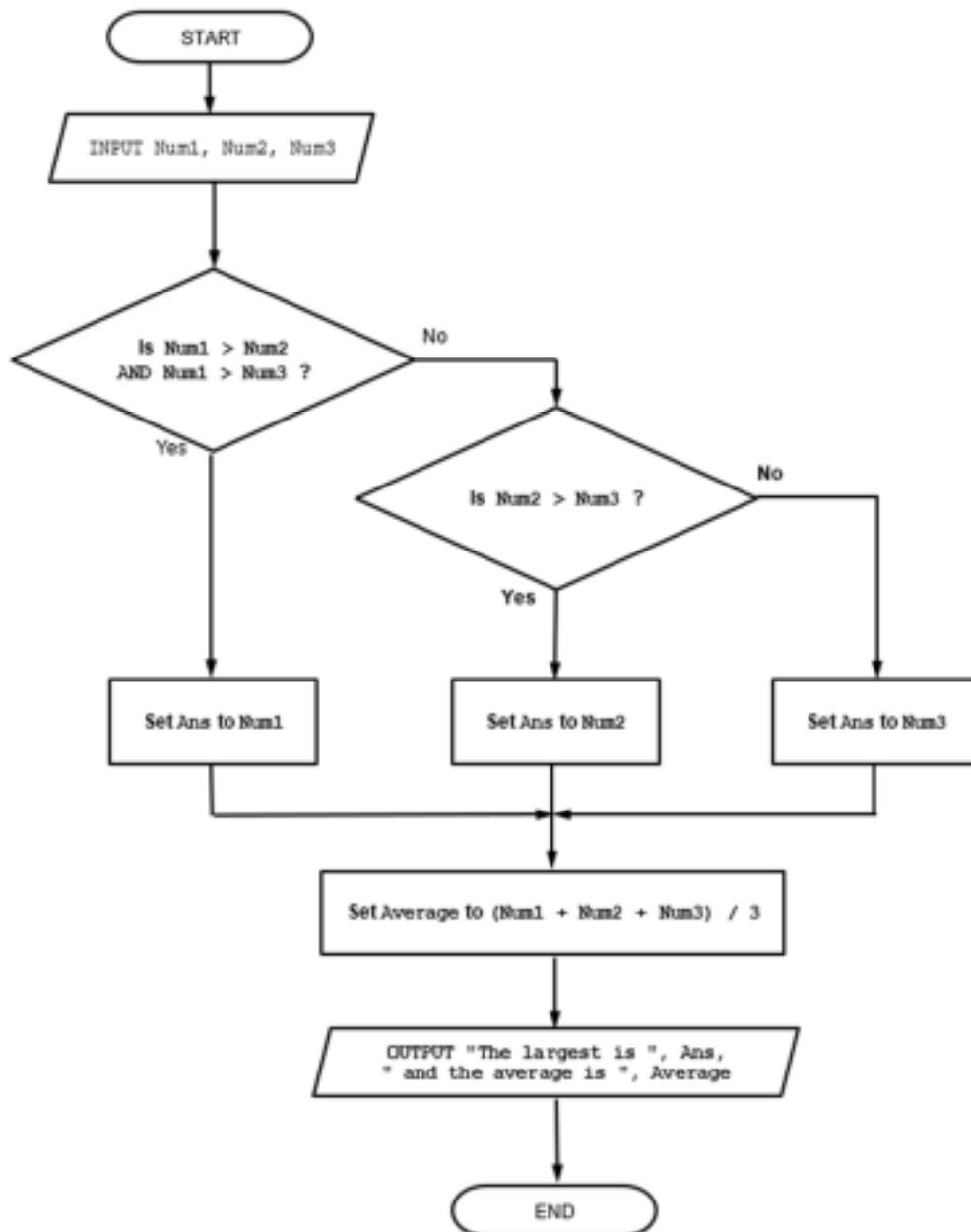
| 1(a)                                                    | <table> <tr> <th>Pseudocode example</th> <th>Selection</th> <th>Iteration</th> <th>Input/Output</th> </tr> <tr> <td>FOR Index ← 1 TO 10<br/>    Data[Index] ← 0<br/>NEXT Index</td> <td></td> <td>✓</td> <td></td> </tr> <tr> <td>WRITEFILE ThisFile,<br/>"****"</td> <td></td> <td></td> <td>✓</td> </tr> <tr> <td>UNTIL Level &gt; 25</td> <td></td> <td>✓</td> <td></td> </tr> <tr> <td>IF Mark &gt; 74 THEN<br/>    READFILE OldFile,<br/>    Data<br/>ENDIF</td> <td>✓</td> <td></td> <td>✓</td> </tr> </table> <p>One mark per row.</p> | Pseudocode example | Selection    | Iteration | Input/Output | FOR Index ← 1 TO 10<br>Data[Index] ← 0<br>NEXT Index |  | ✓ |  | WRITEFILE ThisFile,<br>"****" |  |  | ✓ | UNTIL Level > 25 |  | ✓ |  | IF Mark > 74 THEN<br>READFILE OldFile,<br>Data<br>ENDIF | ✓ |  | ✓ | 4 |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------|-----------|--------------|------------------------------------------------------|--|---|--|-------------------------------|--|--|---|------------------|--|---|--|---------------------------------------------------------|---|--|---|---|
| Pseudocode example                                      | Selection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Iteration          | Input/Output |           |              |                                                      |  |   |  |                               |  |  |   |                  |  |   |  |                                                         |   |  |   |   |
| FOR Index ← 1 TO 10<br>Data[Index] ← 0<br>NEXT Index    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | ✓                  |              |           |              |                                                      |  |   |  |                               |  |  |   |                  |  |   |  |                                                         |   |  |   |   |
| WRITEFILE ThisFile,<br>"****"                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    | ✓            |           |              |                                                      |  |   |  |                               |  |  |   |                  |  |   |  |                                                         |   |  |   |   |
| UNTIL Level > 25                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | ✓                  |              |           |              |                                                      |  |   |  |                               |  |  |   |                  |  |   |  |                                                         |   |  |   |   |
| IF Mark > 74 THEN<br>READFILE OldFile,<br>Data<br>ENDIF | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                    | ✓            |           |              |                                                      |  |   |  |                               |  |  |   |                  |  |   |  |                                                         |   |  |   |   |
| 1(c)                                                    | <p>1 mark for stating a suitable way of documenting:</p> <ul style="list-style-type: none"> <li><u>Identifier table</u><br/>1 mark for giving one piece of information that should be recorded.<br/>examples include:<br/>Explanation of what (each) variable is used for<br/>The purpose of (each) variable<br/>An example of data values stored // Initialisation value</li> </ul>                                                                                                                                                          | 2                  |              |           |              |                                                      |  |   |  |                               |  |  |   |                  |  |   |  |                                                         |   |  |   |   |

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30. M/J 24/P22/Q2/a,b

2(a)

5



Mark points

- 1 Condition for selecting one of the input numbers as largest value
- 2 ... and assign to Ans
- 3 Condition for selecting largest number for all three number input and assigning to Ans
- 4 Average calculated using Num1, Num2 and Num3 and stored in a variable.  
Reject use of DIV
- 5 Output of Ans and average in output symbol / parallelogram

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2(b) | <p><b>Example solutions:</b></p> <pre> Flag ← GetStat() WHILE Flag &lt;&gt; TRUE     FOR Port ← 1 TO 3         CALL Reset(Port)     NEXT Port     Flag ← GetStat() ENDWHILE </pre> <p><b>Alternative:</b></p> <pre> REPEAT     Flag ← GetStat()     IF Flag &lt;&gt; TRUE THEN         FOR Port ← 1 TO 3             CALL Reset(Port)         NEXT Port     ENDIF UNTIL Flag = TRUE </pre> <p><b>One mark per point:</b></p> <ol style="list-style-type: none"> <li>1 (Outer) conditional loop testing <code>Flag</code></li> <li>2 Correct assignment of <code>Flag</code> from <code>GetStat()</code> <b>in a loop</b></li> <li>3 (Inner) loop checking / counting port // Check if <code>Port</code> is different to 4</li> <li>4 ... loop for 3 iterations</li> <li>5 a call to <code>Reset()</code> <b>in a loop</b></li> </ol> | <b>5</b> |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

**31. M/J 24/P22/Q7/a**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 7(a) | <p><b>Examples include:</b></p> <p><b>Module:</b> <code>IdentifyMember()</code><br/> <b>Use:</b> Identifies a club member who has expressed an interest in a given class</p> <p><b>Module:</b> <code>GetMemberPhoneNumber()</code><br/> <b>Use:</b> Gets the mobile phone number of a member</p> <p><b>Module:</b> <code>CreateMessage()</code><br/> <b>Use:</b> Generates a text message to a member</p> <p><b>Module:</b> <code>SendMessage()</code><br/> <b>Use:</b> Sends a text message to a member in the waiting list</p> <p>One mark for name <b>and</b> use</p> <p><b>Note: max 3 marks</b></p> | <b>3</b> |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

**32. M/J 24/P22/Q8/a**

|      |                                                                                                                                                                                                       |          |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 8(a) | <p>One mark for reason, one for benefit</p> <p>Reason: (Program is) easier to design / implement / test / debug / modify</p> <p>Benefit: Easier to check that <b>each stage</b> works as expected</p> | <b>2</b> |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

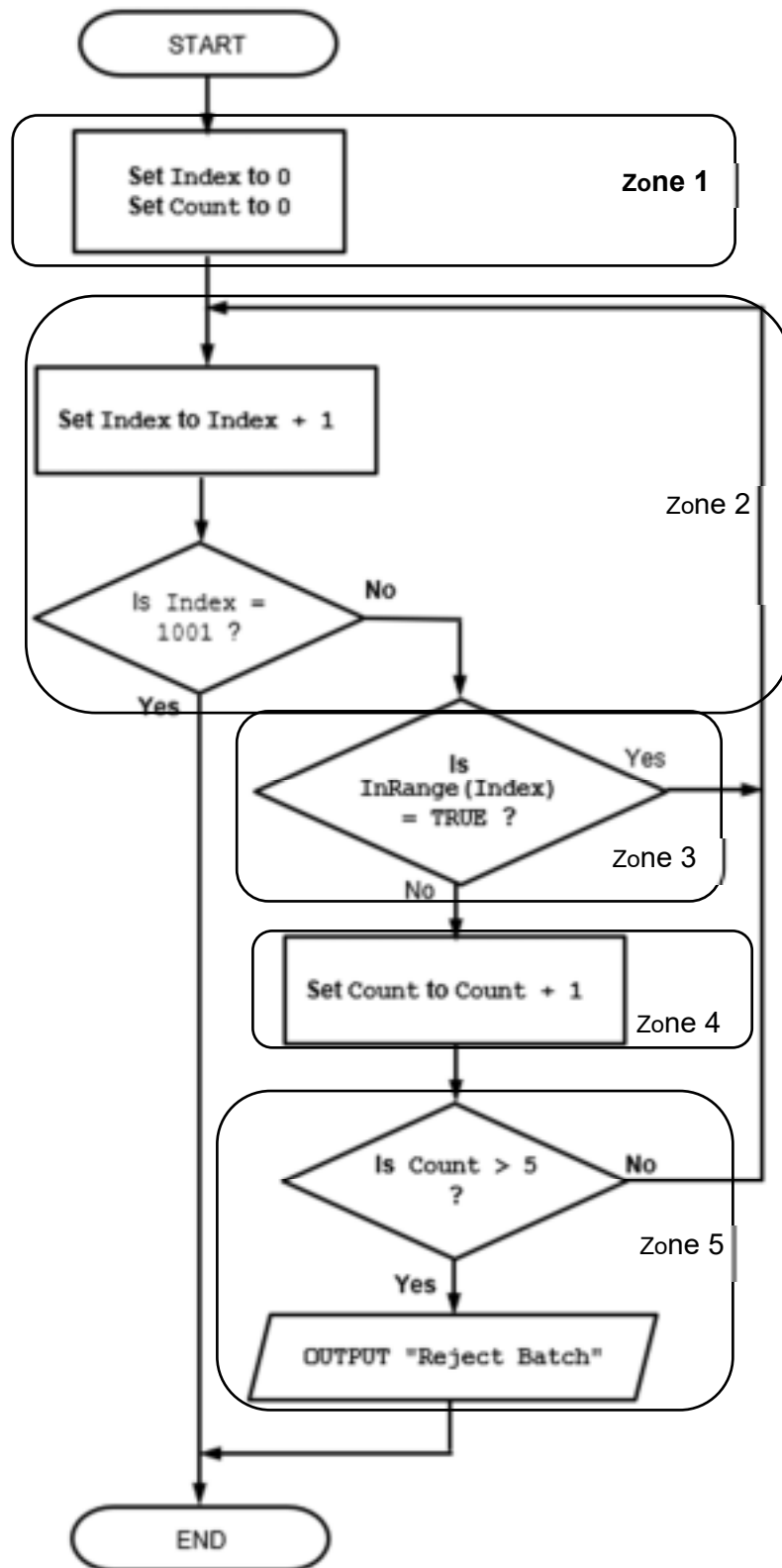
**33. M/J 24/P23/Q2/a,b**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2(a) | <p>One mark per point:</p> <ol style="list-style-type: none"> <li>1. Open the file (in read mode and subsequently close)</li> <li>2. Initialise an index variable (to 1 // 0)</li> <li>3. Repeat (the next three steps) until the end of file is reached</li> <li>4. Read a line from the file (into a string variable)</li> <li>5. Store the line / variable in the array at the index</li> <li>6. Increment the index <b>in a loop</b></li> </ol> <p><b>Note: max 5 marks</b></p> | <b>5</b> |
| 2(b) | <p>One mark per point:</p> <p>Construct: a <u>conditional</u> loop</p> <p>Use: To keep repeating until the <b>end of the file is reached</b></p> <p>ALTERNATIVE:</p> <p>Construct: a <u>selection</u> statement</p> <p>Use: To test / check the value returned by the <code>EOF ( )</code> function</p>                                                                                                                                                                             | <b>2</b> |

**34. M/J 24/P23/Q3/b**

3(b)

5



One mark per zone

**35. O/N 23/P21/Q2/a,b**

|      |                                                                                                                                                                                                                                                                                                                              |          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2(a) | <b>Max 5 marks</b><br><br><b>MP1</b> Set total to <u>zero</u><br><b>MP2</b> Input a number<br><b>MP3</b> Check if number greater than 29 and less than 71<br><b>MP4</b> ... if check is true - add number to total<br><b>MP5</b> Repeat from step 2 99 times // for a total of 100 iterations<br><b>MP6</b> Output the total | <b>5</b> |
| 2(b) | <b>MP1</b> An iterative construct // a (count-controlled) loop<br><b>MP2</b> A selection construct // an IF statement                                                                                                                                                                                                        | <b>2</b> |

**36. O/N 23/P22/Q1/a**

1(a)

4

One mark for each row with appropriate variable name and data type

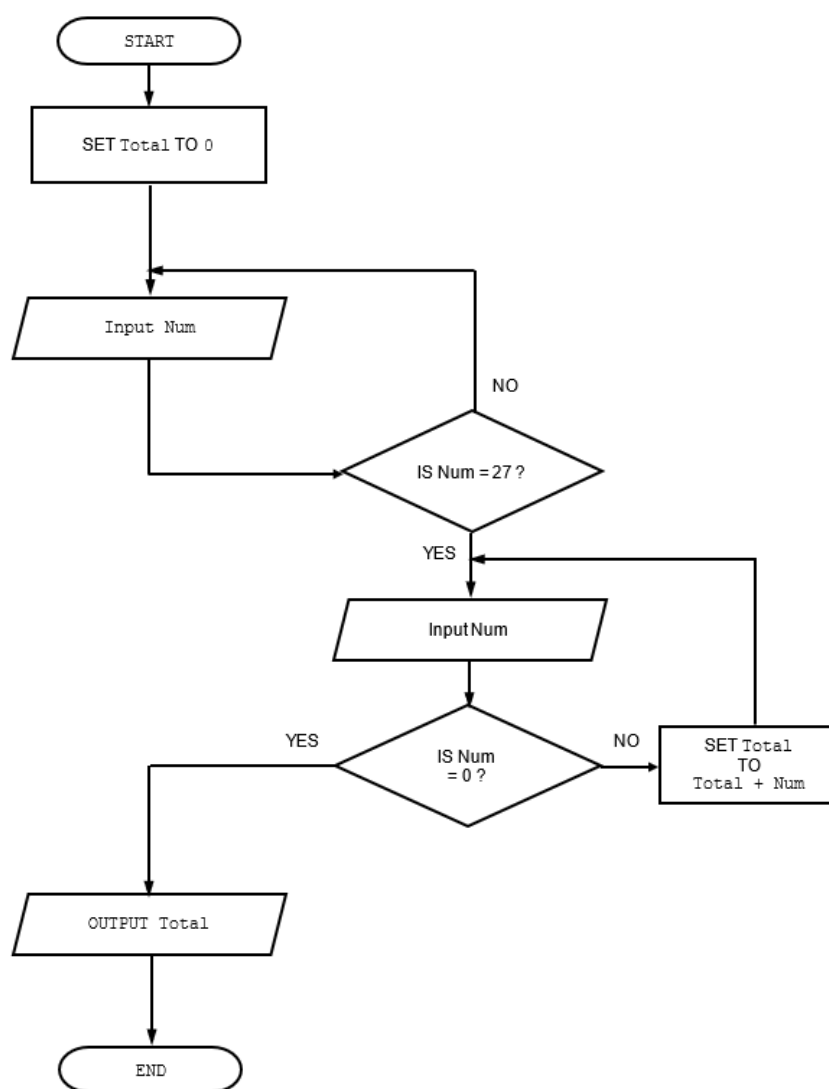
| Example value | Explanation                                        | Variable name | Data type |
|---------------|----------------------------------------------------|---------------|-----------|
| "Mr Khan"     | The name of the customer                           | CustomerName  | STRING    |
| 3             | The number of items in the order                   | NumItems      | INTEGER   |
| TRUE          | A value to indicate whether this is a new customer | NewCustomer   | BOOLEAN   |
| 15.75         | The deposit paid by the customer                   | Deposit       | REAL      |

**37. O/N 23/P22/Q2/a**

2(a)

Example Solution

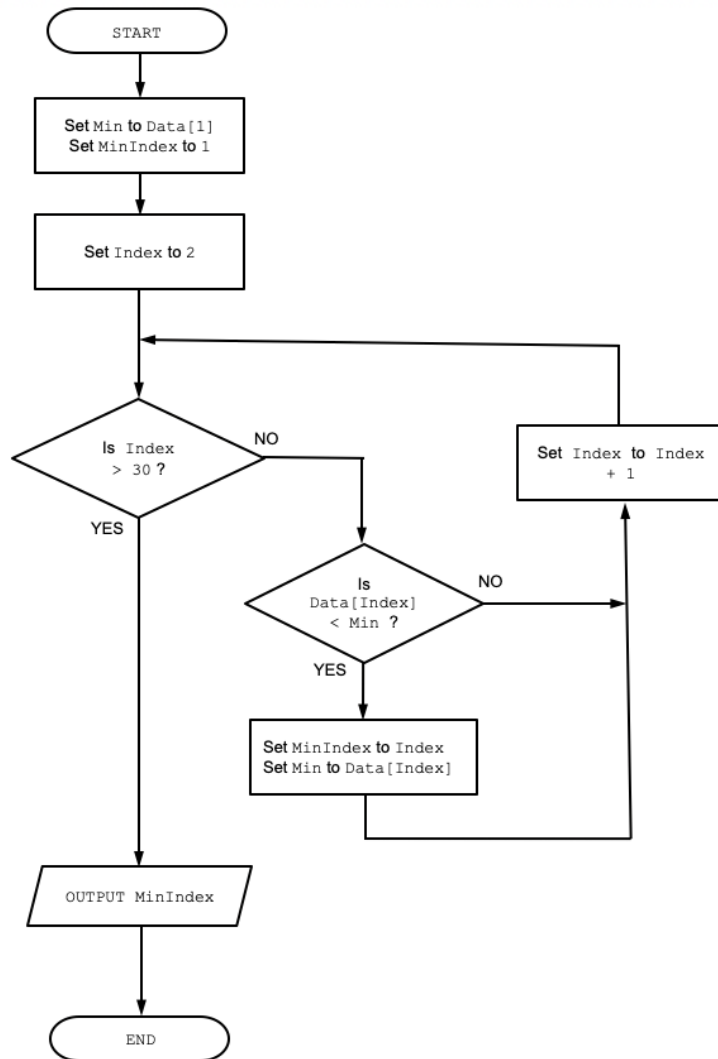
5

**One mark per point:**

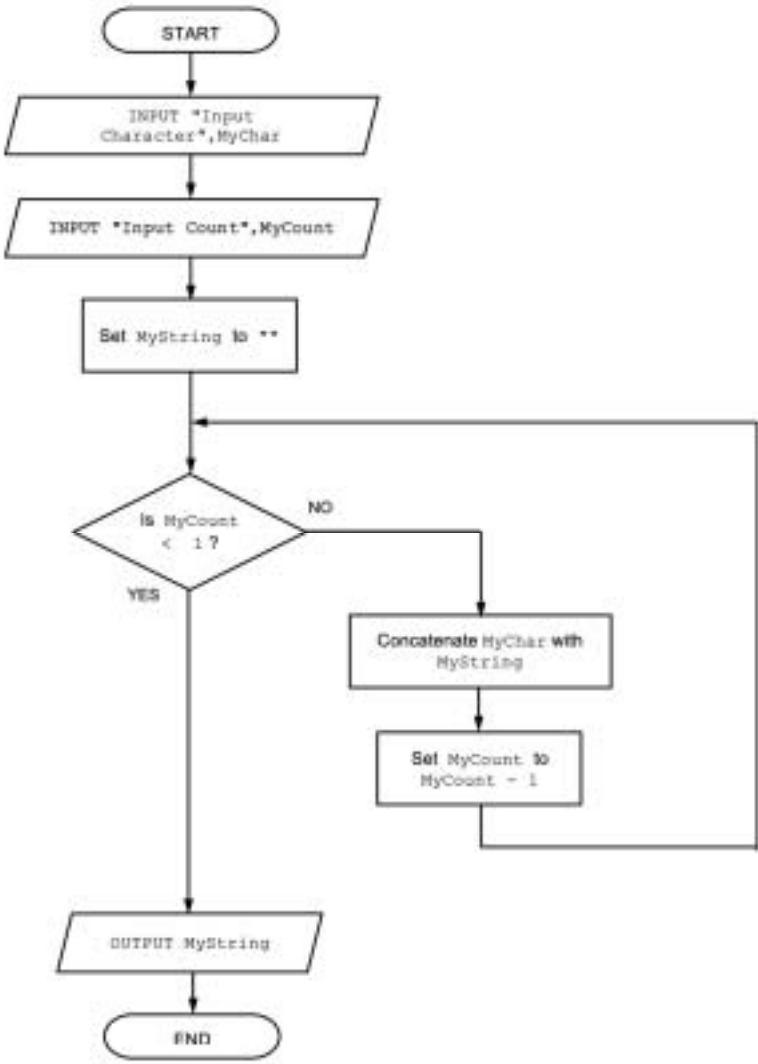
- 1 Initialise `Total` to zero
- 2 Check for only **first** input of 27 **in a loop** then attempt to sum values
- 3 Loop until 0 input
- 4 Sum values **input** (after input of first 27) **in a loop**
- 5 Output `Total` after a reasonable attempt

**38. O/N 23/P23/Q2/a**

2(a)

**5****MP1** Initialise Min to first value in Data **and** MinIndex to 1**MP2** Loop through 29 more values (or 30 values in total)**MP3** Compare element from Data[] with Min**MP4** Set new Min **AND** save MinIndex when element value < Min **in a loop****MP5** Output MinIndex

**39. M/J 23/P21/Q2/a**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 2(a) |  <pre> graph TD     Start([START]) --&gt; InputChar[/INPUT "Input Character", MyChar/]     InputChar --&gt; InputCount[/INPUT "Input Count", MyCount/]     InputCount --&gt; SetString[Set MyString to ""]     SetString --&gt; Decision{Is MyCount &lt; 1?}     Decision -- YES --&gt; Output[/OUTPUT MyString/]     Output --&gt; End([END])     Decision -- NO --&gt; Concat[Concatenate MyChar with MyString]     Concat --&gt; DecCount[Set MyCount to MyCount - 1]     DecCount --&gt; Decision   </pre> <p>One mark per point:</p> <ol style="list-style-type: none"> <li>Both prompts</li> <li>Both inputs using correct identifiers as given in question, <code>MyChar</code> and <code>MyCount</code></li> <li>Initialise <code>MyString</code> to empty string <b>and</b> subsequent output <b>after</b> loop</li> <li>Loop <code>MyCount</code> iterations including decrement <code>MyCount</code></li> <li>Use of Concatenate or equivalent pseudocode statement <b>inside</b> loop</li> </ol> <p><b>Max 4 Marks</b></p> | 4 |
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**40. M/J 23/P21/Q3/b**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |
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| 3(b) | <p>One mark for reference to each:</p> <ol style="list-style-type: none"> <li>1 Reference to the use of constants or variables for the two threshold values of 10 and 100 // Input amount spent (by customer and store in a numeric variable)</li> <li>2 Work out <b>one</b> band that amount maps to</li> <li>3 Work out <b>all</b> bands that amount maps to</li> <li>4 Calculate rounded value of amount / whole number part of amount</li> <li>5 Calculate the points by multiplying the (rounded) amount by the appropriate value for appropriate band /all bands</li> <li>6 Output the number of points</li> </ol> <p><b>Note: Max 5 from available points</b></p> | <b>5</b> |
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**41. M/J 23/P21/Q7/a**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |
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| 7(a) | <p>Examples include:</p> <p>Module: <code>GetOverdueLoan()</code><br/>Use: Identifies an overdue book</p> <p>Module: <code>IdentifyStudent()</code><br/>Use: Identifies a student (with an overdue book)<sup>1</sup></p> <p>Module: <code>GetStudentEmail()</code><br/>Use: Gets the email address of a student with an overdue book</p> <p>Module: <code>CreateEmail()</code><br/>Use: Generates an email to a student with an overdue book</p> <p>Module: <code>SendEmail()</code><br/>Use: Sends an email to a student with an overdue book</p> <p>One mark for name <b>and</b> use</p> <p><b>Note: Max 3 marks</b></p> | <b>3</b> |
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**42. M/J 23/P22/Q1/b**

|      |                                                                                                                                                                                                                                                                                   |          |
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| 1(b) | <p>One mark per bullet point:</p> <ul style="list-style-type: none"> <li>• Indentation</li> <li>• White space</li> <li>• Comments</li> <li>• Sensible / meaningful variable names // use of Camel Case</li> <li>• Capitalised keywords</li> </ul> <p><b>Note: Max 3 marks</b></p> | <b>3</b> |
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**43. M/J 23/P22/Q2/c**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |
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| 2(c) | <p>One mark per array definition bullet:</p> <ul style="list-style-type: none"> <li>• A (1D) array containing 7 elements</li> <li>• of type STRING</li> </ul> <p>One mark per Step:</p> <p>Step1: Assign value "Sunday" to first element, "Monday" to second element etc.</p> <p>Step2: Use the function <code>DAYINDEX()</code> to return / find the day number from <code>MyDoB</code></p> <p>Step3: Use the returned value as the array index / to access the element that contains the name / string</p> <p>Step4: Output the element / name / string</p> <p><b>Note: Max 2 for Array definition, Max 4 for steps</b></p> | <b>6</b> |
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**44. M/J 23/P22/Q3/b**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |
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| 3(b) | <p>One mark per bullet:</p> <ol style="list-style-type: none"> <li>1 Open file in <code>READ</code> mode</li> <li>2 Loop to <code>EOF()</code> // read / process all the lines in file</li> <li>3 Loop will end when return value from <code>AddToQueue()</code> is <code>FALSE</code> / queue is full</li> <li>4 Read a line from the file in a loop</li> <li>5 Pass string to <code>AddToQueue()</code> // <code>AddToQueue()</code> is executed with line as parameter</li> </ol> | <b>5</b> |
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**45. M/J 23/P22/Q7/a**

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 7(a)(i)  | <p>Correct answers include:</p> <ul style="list-style-type: none"> <li>• Information: customer name<br/>Justification: to personalise / address the email</li> <li>• Information: email address<br/>Justification: so that the email can be delivered correctly</li> <li>• Information: product category preference<br/>Justification: to check that the customer would be interested in the product</li> <li>• Information: contact preference<br/>Justification: to check that the customer will accept contact via email</li> <li>• Information: order history<br/>Justification: to send details of product similar to that already bought // to identify frequent shoppers</li> <li>• Information: new product information<br/>Justification: to include information about the new product in the email</li> </ul> <p>One mark for each piece of information <b>and</b> matching justification</p> <p><b>Note: Max 3 marks</b></p> | <b>3</b> |
| 7(a)(ii) | <p>One mark for each piece of information <b>and</b> matching justification:</p> <ul style="list-style-type: none"> <li>• postal address<br/>Justification: sending an email, not a letter</li> <li>• payment details<br/>Justification: Nothing being bought / sold at this stage</li> <li>• order history<br/>Justification: Customer preference used to decide if new product is relevant</li> </ul> <p><b>Note: Max 2 marks</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>2</b> |

**46. M/J 23/P23/Q1/d**

|                                |                                 |              |                |               |          |
|--------------------------------|---------------------------------|--------------|----------------|---------------|----------|
| 1(d)                           | One mark for each of rows 2 - 5 |              |                |               | <b>4</b> |
| <b>Pseudocode statement</b>    |                                 | <b>Input</b> | <b>Process</b> | <b>Output</b> |          |
| INPUT MyChoice                 |                                 | ✓            |                |               |          |
| OUTPUT FirstName & LastName    |                                 |              | ✓              | ✓             |          |
| WRITEFILE OutputFile, TextLine |                                 |              |                | ✓             |          |
| READFILE MyFile, TextLine      |                                 | ✓            |                |               |          |
| Result ← SQRT (NextNum)        |                                 |              | ✓              |               |          |

**47. M/J 23/P23/Q3/b(ii)**

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 3(b)(ii) | <p><b>Max 5 marks</b></p> <p>MP1 Open the text file in WRITE mode</p> <p>MP2 Check there is a value on the stack</p> <p>MP3 POP value ....</p> <p>MP4 Write value to the text file</p> <p>MP5 Repeat from Step 2 // loop referencing the stack items</p> <p>Alternative solution: Not using POP primitive</p> <p>MP1 Open the text file in WRITE mode</p> <p>MP2 Check there is a value on the stack</p> <p>MP3 Read value from ToS location</p> <p>MP4 Write the value to the text file – Must some attempt at ‘the value’<br/>NOT ‘<u>all</u> the values’</p> <p>MP5 Decrement ToS</p> <p>MP6 Repeat from step 2 // loop referencing the stack items</p> | <b>5</b> |
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**48. M/J 23/P23/Q7/a**

|          |                                                                                                                                                                                                                                                                                        |          |
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| 7(a)(i)  | <ul style="list-style-type: none"> <li>To filter out information (that is not necessary to solve the problem) // to include only essential information</li> </ul>                                                                                                                      | <b>1</b> |
| 7(a)(ii) | <p><b>Required:</b></p> <p>Student : Student name / email (address)</p> <p>Loan: Return/Issue date</p> <p>Book: Book title</p> <p><b>Not Required:</b></p> <p>Student: Home address / DoB / tutor / subject choices</p> <p>Book: Library location / category / author / book title</p> | <b>2</b> |

|           |                                                                                                                                                                                                                                                                                                                                                          |          |
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| 7(a)(iii) | <p><b>Max 2 marks</b></p> <p>Examples could include:</p> <ul style="list-style-type: none"> <li>• Clear the loan // indicate that the book has been returned // update loan history</li> <li>• Take the student off 'block' // allow the student to borrow further books</li> <li>• Send acknowledgement to the student when book is returned</li> </ul> | <b>2</b> |
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**49. O/N 22/P21/Q2/a,b**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2(a) | <p>One mark for Explanation:</p> <ul style="list-style-type: none"> <li>• Abstraction is used to filter out information / data that is not necessary for the task</li> </ul> <p>Or the opposite:</p> <ul style="list-style-type: none"> <li>• To keep only information / data that is necessary for the task</li> </ul> <p>One mark for each <b>TWO</b> data items (not dependent on 'Explanation'):</p> <p>Items include:</p> <ul style="list-style-type: none"> <li>• Car details: ID, Car Registration, car type etc</li> <li>• Customer details: ID, name, address, licence details etc</li> <li>• Start date (of hire)</li> <li>• Return date / Number of days (of hire)</li> <li>• Cost of hire</li> </ul> | <b>3</b> |
| 2(b) | <p>One mark for each (<b>Max 2</b>)</p> <p>Examples include:</p> <ol style="list-style-type: none"> <li>1 Input customer details</li> <li>2 Input car details</li> <li>3 Input payment details</li> <li>4 Create hire / start hire</li> <li>5 Return car / end hire</li> <li>6 Change / check car status (hired / available / written-off)</li> <li>7 Cancel hire</li> <li>8 Process payment / calculate hire cost</li> </ol>                                                                                                                                                                                                                                                                                    | <b>2</b> |

**50. O/N 22/P21/Q3**

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 3 | <p>One mark per point (<b>Max 5</b>):</p> <ol style="list-style-type: none"> <li>1 Declare a variable / an integer <code>Max</code></li> <li>2 Assign value of first / any element to <code>Max</code></li> <li>3 Set up a loop to repeat 200 times / from start to end of array</li> <li>4 Use the loop counter as the array index</li> <li>5 If value of current element is greater than <code>Max</code>...</li> <li>6 ...then assign value to <code>Max</code></li> <li>7 After the loop, Output <code>Max</code></li> </ol> | <b>5</b> |
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**51. O/N 22/P22/Q2/a**

|           |                                                                                                                                                                                                                                                                                                                                                                                       |          |
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| 2(a)(i)   | <p>One mark per step (or equivalent):</p> <ol style="list-style-type: none"> <li>1 Open file in <u>APPEND</u> mode (and subsequent Close)</li> <li>2 Prompt and Input a student name and mark</li> <li>3 If mark greater than or equal to 20 jump to step 5</li> <li>4 Write only the name to the file</li> <li>5 Repeat from Step 2 for 35 times / the number of students</li> </ol> | <b>5</b> |
| 2(a)(ii)  | <p>Data in a file is saved after the computer is switched off / stored permanently // no need to re-enter the data when the program is re-run</p>                                                                                                                                                                                                                                     | <b>1</b> |
| 2(a)(iii) | <p>Example answer:</p> <p>So that existing file data is not overwritten.</p>                                                                                                                                                                                                                                                                                                          | <b>1</b> |

**52. O/N 22/P22/Q4/a**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
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| 4(a) | <p>One mark per point:</p> <ol style="list-style-type: none"> <li>1 Input UserID <b>and</b> use of GetAverage() <b>and</b> assignment</li> <li>2 Initialisation of Total to zero <b>and</b> Index to 4</li> <li>3 Conditional loop with Index from 4 to 6</li> <li>4 Assignment of Last from element SameMonth[Index] <b>in a loop</b></li> <li>5 Structure: IF...THEN...ELSE...ENDIF <b>in a loop</b></li> <li>6 Correct assignments <b>and</b> final call to Update() <b>after the loop</b></li> </ol> <pre> INPUT UserID Average ← GetAverage(UserID) Total ← 0 Index ← 4  WHILE Index &lt; 7 // REPEAT     Last ← SameMonth[Index]     IF Average &gt; Last THEN         Total ← Total + Average     ELSE         Total ← Total + Last     ENDIF     Index ← Index + 1 ENDWHILE // UNTIL Index = 7  CALL Update(UserID, Total) </pre> <p><u>Alternative solution using FOR loop:</u></p> <p>One mark per point FOR loop solution:</p> <ol style="list-style-type: none"> <li>1 Input UserID <b>and</b> use of GetAverage() <b>and</b> assignment</li> <li>2 Initialisation of Total to zero</li> <li>3 loop Index from 4 to 6</li> <li>4 Assignment of Last from array SameMonth <b>in a loop</b></li> <li>5 Comparison IF...THEN...ELSE...ENDIF <b>in a loop</b></li> <li>6 Appropriate assignments <b>in a loop AND</b> final call to Update() <b>after the loop</b></li> </ol> <pre> INPUT UserID Average ← GetAverage(UserID) Total ← 0 FOR Index ← 4 TO 6     Last ← SameMonth[Index]     IF Average &gt; Last THEN         Total ← Total + Average     ELSE         Total ← Total + Last     ENDIF NEXT Index  CALL Update(UserID, Total) </pre> | 6 |
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**53. O/N 22/P22/Q5**

|   |                                                                                                                                                                                                                                                                                               |   |
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| 5 | <p>One mark per IF...THEN...ENDIF clause:</p> <p>1 IF A AND B AND C THEN<br/>CALL Sub1()<br/>ENDIF</p> <p>2 IF (A AND B) AND NOT C THEN<br/>CALL Sub2()<br/>ENDIF</p> <p>3 IF (NOT A) AND (NOT C) THEN<br/>CALL Sub3()<br/>ENDIF</p> <p>4 IF (NOT A) AND C THEN<br/>CALL Sub4()<br/>ENDIF</p> | 4 |
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**54. O/N 22/P23/Q2**

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
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| 2 | <p>One mark for name <b>and</b> two marks for use (<b>Max 3 in total</b>):</p> <p>Examples include:</p> <p>Module: SelectCharity()<br/>Use: Allows the user to choose a particular charity</p> <p>Module: SpecifyAmountAndType()<br/>Use: Allows the user to specify a single or regular payment</p> <p>Module: MakePayment()<br/>Use: Make payment to the charity</p> <p>Module: ValidatePayment()<br/>Use: Validate payment details (by accessing bank computer)</p> <p>Module: AddBankAccountDetails() / AddPaymentDetails()<br/>Use: Allows the user to add bank account information that donation to be taken from</p> <p>Module: AddDonorDetails()<br/>Use: Allows user to add details such as name and contact details</p> | 3 |
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**55. O/N 22/P23/Q3**

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 3 | <p>One mark per point, for example:</p> <ol style="list-style-type: none"> <li>1 Declare two (REAL) variables for the two sum values <b>AND</b> initialise both to zero</li> <li>2 Prompt <b>AND</b> Input a number</li> <li>3 If number greater than zero add to positive sum and If number less than zero add to negative sum</li> <li>4 Repeat from step 2 if number not zero</li> <li>5 After loop the Output SumPos and SumNeg</li> </ol> | 5 |
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**56. O/N 22/P23/Q5/c**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
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| 5(c) | <p>One mark for reason<br/>One mark for each example (<b>Max 2</b>)</p> <p>Reason:</p> <ul style="list-style-type: none"> <li>• Makes the code easier to understand // Describes the purpose of the identifier // Makes the code easier to debug/test/maintain</li> </ul> <p>Further examples include:</p> <ul style="list-style-type: none"> <li>• White space</li> <li>• Indentation</li> <li>• Keywords in capitals</li> <li>• Comments</li> <li>• Local variables // parameters</li> </ul> | 3 |
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**57. O/N 22/P23/Q7/b(ii)**

|          |                                                                                                                                                                                                                                                                                                                                                                             |   |
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| 7(b)(ii) | <p>One mark per point (<b>Max 3</b>):</p> <ol style="list-style-type: none"> <li>1. Loop through 500 elements (while error number not found)</li> <li>2. Compare ErrCode for current element with the error number</li> <li>3. If same, set element value to 999 (and terminate loop)</li> <li>4. ... and call SortArrays() (to move 999 to the end) – once only</li> </ol> | 3 |
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**58. O/N 21/P21/Q1/c**

| 1(c)(i)                     | <table border="1"> <thead> <tr> <th>Information</th><th>Essential</th><th>Not essential</th></tr> </thead> <tbody> <tr> <td>Departure time</td><td>✓</td><td></td></tr> <tr> <td>Flight Number</td><td></td><td>✓</td></tr> <tr> <td>Departure airport</td><td>✓</td><td></td></tr> <tr> <td>Aircraft type</td><td></td><td>✓</td></tr> <tr> <td>Ticket price</td><td>✓</td><td></td></tr> <tr> <td>Number of seats in aircraft</td><td></td><td>✓</td></tr> </tbody> </table> <p>One mark for two rows correct<br/>Two mark for four rows correct<br/>Three mark for all rows correct</p> | Information   | Essential | Not essential | Departure time | ✓ |  | Flight Number |  | ✓ | Departure airport | ✓ |  | Aircraft type |  | ✓ | Ticket price | ✓ |  | Number of seats in aircraft |  | ✓ | 3 |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------|---------------|----------------|---|--|---------------|--|---|-------------------|---|--|---------------|--|---|--------------|---|--|-----------------------------|--|---|---|
| Information                 | Essential                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Not essential |           |               |                |   |  |               |  |   |                   |   |  |               |  |   |              |   |  |                             |  |   |   |
| Departure time              | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |           |               |                |   |  |               |  |   |                   |   |  |               |  |   |              |   |  |                             |  |   |   |
| Flight Number               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ✓             |           |               |                |   |  |               |  |   |                   |   |  |               |  |   |              |   |  |                             |  |   |   |
| Departure airport           | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |           |               |                |   |  |               |  |   |                   |   |  |               |  |   |              |   |  |                             |  |   |   |
| Aircraft type               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ✓             |           |               |                |   |  |               |  |   |                   |   |  |               |  |   |              |   |  |                             |  |   |   |
| Ticket price                | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |           |               |                |   |  |               |  |   |                   |   |  |               |  |   |              |   |  |                             |  |   |   |
| Number of seats in aircraft |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ✓             |           |               |                |   |  |               |  |   |                   |   |  |               |  |   |              |   |  |                             |  |   |   |

|           |                                                                                                                                                                                                                                                                                                                     |          |
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| 1(c)(ii)  | <p>One mark for technique and one for benefit, <b>Max 1</b> mark for 'Benefit'</p> <p>Technique: Abstraction</p> <p>Benefit:</p> <ul style="list-style-type: none"> <li>• The solution is simplified so easier / quicker to design / implement</li> <li>• The system is tailored to the need of the user</li> </ul> | <b>2</b> |
| 1(c)(iii) | <p>Answers include:</p> <ul style="list-style-type: none"> <li>• Destination / arrival airport</li> <li>• Arrival time / flight duration</li> <li>• Date of flight</li> <li>• Seat number</li> <li>• Seat availability</li> </ul> <p><b>Max 2</b> marks</p>                                                         | <b>2</b> |

**59. O/N 21/P21/Q2/a,b**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2(a) | <p>One mark for reference to:</p> <ol style="list-style-type: none"> <li>1 The use a variable as an index to the array</li> <li>2 A loop to iterate through the array</li> <li>3 An Inner loop (with a reducing range)</li> <li>4 Test if current element is greater than next element</li> <li>5 if so then swap elements</li> <li>6 Description of swap</li> <li>7 Attempt at efficient algorithm</li> </ol> <p><b>Max 6</b> marks</p>                                                                                                                                                                                                                 | <b>6</b> |
| 2(b) | <pre> Count ← 1 Flag ← FALSE WHILE Flag = FALSE AND Count &lt;= 5     CALL ReBoot()     Count ← Count + 1     Flag ← Check() ENDWHILE  IF Flag = FALSE THEN     CALL Alert(27) ENDIF </pre> <p>One mark per point:</p> <ol style="list-style-type: none"> <li>1 Initialisation of Count <b>AND</b> Flag</li> <li>2 WHILE ... ENDWHILE // REPEAT ... UNTIL loop</li> <li>3 ... including both conditions</li> <li>4 Call ReBoot() <b>AND</b> increment Count <b>inside the loop</b></li> <li>5 Assign return value from Check() to Flag <b>inside the loop</b></li> <li>6 Final test of Flag <b>AND</b> call to Alert(27) <b>not in a loop</b></li> </ol> | <b>6</b> |

**60. O/N 21/P22/Q1/a**

|      |                                                                                                                                                                                                                                                                                                                                               |          |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1(a) | <p>The process involves:</p> <ol style="list-style-type: none"><li>1 Breaking down a problem / task into sub problems / steps / smaller parts</li><li>2 In order to explain / understand // easier to solve the problem</li><li>3 Leading to the concept of program modules // assigning problem parts to teams</li></ol> <p><b>Max 2</b></p> | <b>2</b> |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

**61. O/N 21/P22/Q2/a**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 2(a) | <div><pre>graph TD; START([START]) --&gt; SetPosCount[Set PosCount, NegCount to 0]; SetPosCount --&gt; SetPosTotal[Set PosTotal, NegTotal to 0]; SetPosTotal --&gt; InputNextNum[/INPUT NextNum/]; InputNextNum --&gt; IsZero{Is NextNum = 0?}; IsZero -- YES --&gt; OutputPosCount[/OUTPUT PosCount, NegCount/]; OutputPosCount --&gt; OutputPosTotal[/OUTPUT PosTotal, NegTotal/]; OutputPosTotal --&gt; END([END]); IsZero -- NO --&gt; IsPos{Is NextNum &gt; 0?}; IsPos -- YES --&gt; IncPosCount[Increment PosCount]; IncPosCount --&gt; SetPosTotalAdd[Set PosTotal to PosTotal + NextNum]; SetPosTotalAdd --&gt; Merge1(( )); IsPos -- NO --&gt; IncNegCount[Increment NegCount]; IncNegCount --&gt; SetNegTotalAdd[Set NegTotal to NegTotal + NextNum]; SetNegTotalAdd --&gt; Merge1; Merge1 --&gt; InputNextNum;</pre></div> <p>One mark for each outlined group.</p> <p>Note:</p> <ul style="list-style-type: none"><li>• Sum and increment steps (bottom-right rectangles) may be in reverse order in which case sum group will have two output lines</li><li>• Max 4 for non-working solution</li></ul> | 5 |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

62. O/N 21/P22/Q4/b

|      |                                                                                                                                                                                               |   |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 4(b) | <p>One mark per point:</p> <ul style="list-style-type: none"><li>• Meaningful variable names</li><li>• Indentation / white space / blank lines</li><li>• Capitalisation of keywords</li></ul> | 3 |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

**63. O/N 21/P23/Q1/c**

| 1(c)(i)                                                                                            | <table><tr><th>Information</th><th>Essential</th><th>Not essential</th></tr><tr><td>Departure time</td><td>✓</td><td></td></tr><tr><td>Flight Number</td><td></td><td>✓</td></tr><tr><td>Departure airport</td><td>✓</td><td></td></tr><tr><td>Aircraft type</td><td></td><td>✓</td></tr><tr><td>Ticket price</td><td>✓</td><td></td></tr><tr><td>Number of seats in aircraft</td><td></td><td>✓</td></tr></table> | Information | Essential     | Not essential | Departure time | ✓ |  | Flight Number |  | ✓ | Departure airport | ✓ |  | Aircraft type |  | ✓ | Ticket price | ✓ |  | Number of seats in aircraft |  | ✓ | 3 |
|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------|---------------|----------------|---|--|---------------|--|---|-------------------|---|--|---------------|--|---|--------------|---|--|-----------------------------|--|---|---|
|                                                                                                    | Information                                                                                                                                                                                                                                                                                                                                                                                                        | Essential   | Not essential |               |                |   |  |               |  |   |                   |   |  |               |  |   |              |   |  |                             |  |   |   |
|                                                                                                    | Departure time                                                                                                                                                                                                                                                                                                                                                                                                     | ✓           |               |               |                |   |  |               |  |   |                   |   |  |               |  |   |              |   |  |                             |  |   |   |
|                                                                                                    | Flight Number                                                                                                                                                                                                                                                                                                                                                                                                      |             | ✓             |               |                |   |  |               |  |   |                   |   |  |               |  |   |              |   |  |                             |  |   |   |
|                                                                                                    | Departure airport                                                                                                                                                                                                                                                                                                                                                                                                  | ✓           |               |               |                |   |  |               |  |   |                   |   |  |               |  |   |              |   |  |                             |  |   |   |
|                                                                                                    | Aircraft type                                                                                                                                                                                                                                                                                                                                                                                                      |             | ✓             |               |                |   |  |               |  |   |                   |   |  |               |  |   |              |   |  |                             |  |   |   |
|                                                                                                    | Ticket price                                                                                                                                                                                                                                                                                                                                                                                                       | ✓           |               |               |                |   |  |               |  |   |                   |   |  |               |  |   |              |   |  |                             |  |   |   |
|                                                                                                    | Number of seats in aircraft                                                                                                                                                                                                                                                                                                                                                                                        |             | ✓             |               |                |   |  |               |  |   |                   |   |  |               |  |   |              |   |  |                             |  |   |   |
| One mark for two rows correct<br>Two mark for four rows correct<br>Three mark for all rows correct |                                                                                                                                                                                                                                                                                                                                                                                                                    |             |               |               |                |   |  |               |  |   |                   |   |  |               |  |   |              |   |  |                             |  |   |   |

|          |                                                                                                                                                                                                                                                                                                           |   |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 1(c)(ii) | One mark for technique and one for benefit, <b>Max 1</b> mark for 'Benefit'<br><br>Technique: Abstraction<br><br>Benefit: <ul style="list-style-type: none"><li>• The solution is simplified so easier / quicker to design / implement</li><li>• The system is tailored to the need of the user</li></ul> | 2 |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

|           |                                                                                                                                                                                                                                            |   |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 1(c)(iii) | Answers include: <ul style="list-style-type: none"><li>• Destination / arrival airport</li><li>• Arrival time / flight duration</li><li>• Date of flight</li><li>• Seat number</li><li>• Seat availability</li></ul><br><b>Max 2</b> marks | 2 |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

**64. O/N 21/P23/Q2/a,b**

|      |                                                        |          |
|------|--------------------------------------------------------|----------|
| 2(a) | One mark for reference to:                             | <b>6</b> |
|      | 1 The use a variable as an index to the array          |          |
|      | 2 A loop to iterate through the array                  |          |
|      | 3 An Inner loop (with a reducing range)                |          |
|      | 4 Test if current element is greater than next element |          |
|      | 5 if so then swap elements                             |          |
|      | 6 Description of swap                                  |          |
|      | 7 Attempt at efficient algorithm                       |          |
|      | <b>Max 6</b> marks                                     |          |

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 2(b) | <pre> Count ← 1 Flag ← FALSE WHILE Flag = FALSE AND Count ≤ 5     CALL ReBoot()     Count ← Count + 1     Flag ← Check() ENDWHILE  IF Flag = FALSE THEN     CALL Alert(27) ENDIF </pre> <p>One mark per point:</p> <ol style="list-style-type: none"> <li>1 Initialisation of Count <b>AND</b> Flag</li> <li>2 WHILE ... ENDWHILE // REPEAT ... UNTIL loop</li> <li>3 ... including both conditions</li> <li>4 Call ReBoot() <b>AND</b> increment Count <b>inside the loop</b></li> <li>5 Assign return value from Check() to Flag <b>inside the loop</b></li> <li>6 Final test of Flag <b>AND</b> call to Alert(27) <b>not in a loop</b></li> </ol> | 6 |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

## 65. M/J 21/P21/Q1/a

1(a)

| Example value | Explanation                                                                                         | Variable name       | Data type      |
|---------------|-----------------------------------------------------------------------------------------------------|---------------------|----------------|
| "Wong"        | The preferred name of the member joining the football club                                          | <b>MemberName</b>   | <b>STRING</b>  |
| FALSE         | A value to indicate whether an existing member of the club lives at the same address                | <b>FamilyMember</b> | <b>BOOLEAN</b> |
| 19/02/1983    | When the member joined the football club                                                            | <b>StartDate</b>    | <b>DATE</b>    |
| 1345          | The number of points a member has earned. Members of the club earn points for different activities. | <b>Points</b>       | <b>INTEGER</b> |

One mark for each appropriate variable name plus data type

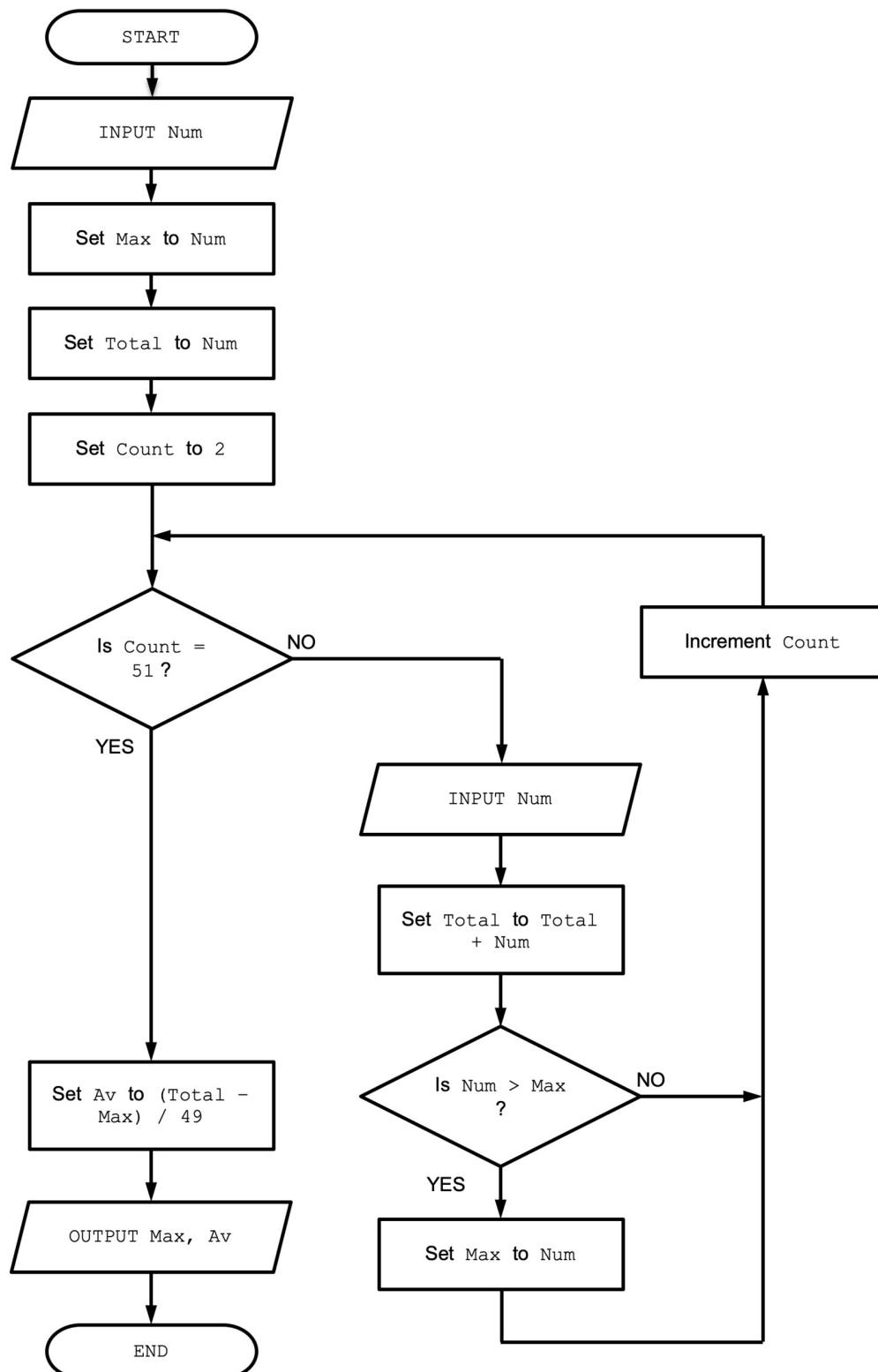
4

## 66. M/J 21/P21/Q2/c

2(c)

Example solution:

6



One mark for each functional group as listed below:

|      |                                                                                                                                                                                                                                                                                                                                                                                                             |  |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 2(c) | <p>Explanation of mark points:</p> <ol style="list-style-type: none"> <li>1 Initialise MAX to first value input</li> <li>2 Set Total to zero</li> <li>3 Input 49 more values (or 50 values in total)</li> <li>4 Sum all values input</li> <li>5 Set new MAX when Input value &gt; MAX <b>in a loop</b></li> <li>6 Sum all but largest (or subtract MAX from total), calculate and output average</li> </ol> |  |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

**67. M/J 21/P21/Q7/b**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 7(b) | <p>Marks awarded for any reference to each of the following steps of the algorithm:</p> <ol style="list-style-type: none"> <li>1 Mention of variable for use as array index</li> <li>2 Use of a loop (to check through the array)</li> <li>3 If word is the same as the current array element then return FALSE / set flag</li> <li>4 If word not already in array, loop to find unused element (<i>second loop</i>)</li> <li>5 Store word in unused element and return TRUE, otherwise return FALSE</li> </ol> <p>VARIATION:</p> <ol style="list-style-type: none"> <li>1 Mention of variable for use as array index</li> <li>2 Use of a loop (to check through the array)</li> <li>3 Save index of (first) unused element found</li> <li>4 If word is the same as the current array element then return FALSE / set flag</li> <li>5 If word not already in array <b>and</b> unused element available, store word in unused element and return TRUE otherwise return FALSE</li> </ol> <p><b>Note: Max 4 marks</b></p> | <b>4</b> |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

**68. M/J 21/P22/Q1/b**

| 1(b)             | <table><tr><th>Statement</th><th>Selection</th><th>Assignment</th><th>Iteration</th></tr><tr><td>Index ← Index + 1</td><td></td><td>✓</td><td></td></tr><tr><td>IF Modified = TRUE THEN</td><td>✓</td><td></td><td></td></tr><tr><td>ENDWHILE</td><td></td><td></td><td>✓</td></tr></table> | Statement | Selection  | Assignment | Iteration | Index ← Index + 1 |  | ✓ |  | IF Modified = TRUE THEN | ✓ |  |  | ENDWHILE |  |  | ✓ | 3 |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------|------------|-----------|-------------------|--|---|--|-------------------------|---|--|--|----------|--|--|---|---|
|                  | Statement                                                                                                                                                                                                                                                                                   | Selection | Assignment | Iteration  |           |                   |  |   |  |                         |   |  |  |          |  |  |   |   |
|                  | Index ← Index + 1                                                                                                                                                                                                                                                                           |           | ✓          |            |           |                   |  |   |  |                         |   |  |  |          |  |  |   |   |
|                  | IF Modified = TRUE THEN                                                                                                                                                                                                                                                                     | ✓         |            |            |           |                   |  |   |  |                         |   |  |  |          |  |  |   |   |
| ENDWHILE         |                                                                                                                                                                                                                                                                                             |           | ✓          |            |           |                   |  |   |  |                         |   |  |  |          |  |  |   |   |
| One mark per row |                                                                                                                                                                                                                                                                                             |           |            |            |           |                   |  |   |  |                         |   |  |  |          |  |  |   |   |

|                        |                 |                           |                  |          |
|------------------------|-----------------|---------------------------|------------------|----------|
| 1(a)(i)                | <b>Variable</b> | <b>Example data value</b> | <b>Data type</b> | <b>4</b> |
|                        | Name            | "Catherine"               | <b>STRING</b>    |          |
|                        | Index           | 100                       | <b>INTEGER</b>   |          |
|                        | Modified        | FALSE                     | <b>BOOLEAN</b>   |          |
|                        | Holiday         | 25/12/2020                | <b>DATE</b>      |          |
| One mark per data type |                 |                           |                  |          |

**69. M/J 21/P22/Q2/b**

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2(b)(i)   | <p>Answers include:</p> <ul style="list-style-type: none"> <li>• User ID / Username</li> <li>• Book ID</li> <li>• Date of loan / return date</li> </ul> <p>One mark for 1 correct<br/>Two marks for all 3 correct</p> <p><b>Note: Max 2 marks</b></p>                                                                                                                                                                                                                                                                                                          | <b>2</b> |
| 2(b)(ii)  | <p>Many examples but must be data that is NOT required for a loan, but which COULD be required somewhere by the library system.</p> <p>Note: must be data relating to users, books or loans</p> <p>Answers include:</p> <ul style="list-style-type: none"> <li>• Users name / address / phone number / DOB</li> <li>• Book title / author / publisher / library rack number / ISBN number / price</li> <li>• Date of loan / return date (if not already given in part (i))</li> <li>• The length of the loan (assumed to be the same for all books)</li> </ul> | <b>1</b> |
| 2(b)(iii) | <p>Many examples including:</p> <ul style="list-style-type: none"> <li>• Create loan / borrow book</li> <li>• Return book</li> <li>• Send letter / email / contact a user ref an overdue book</li> <li>• View the loan history for a given book</li> <li>• View the loan history for a given user</li> </ul> <p>One mark for each</p> <p><b>Note: Max 2 marks</b></p>                                                                                                                                                                                          | <b>2</b> |

**70. M/J 21/P22/Q4**

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>4</b> | <p>Marks awarded for a description of each of the following steps of the algorithm:</p> <ol style="list-style-type: none"> <li>1 Reference variables for <code>Count</code> of students and <code>Total</code> marks</li> <li>2 Loop through all students (<code>Count</code>)</li> <li>3 Input individual mark (<b>in loop</b>)</li> <li>4 Compare mark with threshold / boundary values to determine grade (<b>in loop</b>)</li> <li>5 Output the grade for a student (<b>in loop</b>)</li> <li>6 Maintain a <code>Total</code> (and <code>Count</code> if required) (<b>in loop</b>)</li> <li>7 Calculate average by dividing <code>Total</code> by <code>Count</code> <b>and</b> Output (<b>after loop</b>)</li> </ol> <p><b>Note: Max 6 marks</b></p> | <b>6</b> |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

**71. M/J 21/P23/Q1/a**

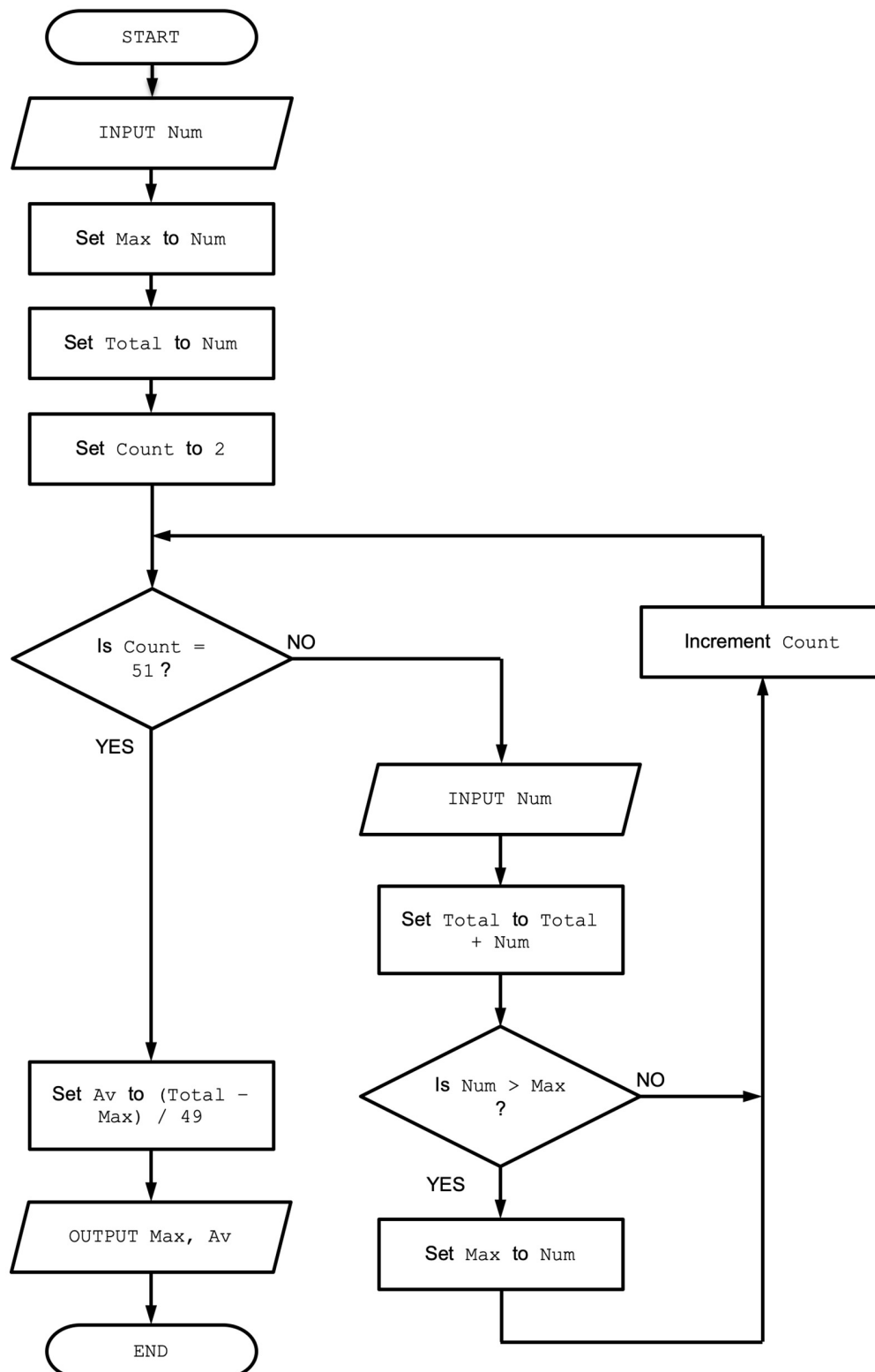
| 1(a)                                                       | <table><tr><th>Example value</th><th>Explanation</th><th>Variable name</th><th>Data type</th></tr><tr><td>"Wong"</td><td>The preferred name of the member joining the football club</td><td>MemberName</td><td>STRING</td></tr><tr><td>FALSE</td><td>A value to indicate whether an existing member of the club lives at the same address</td><td>FamilyMember</td><td>BOOLEAN</td></tr><tr><td>19/02/1983</td><td>When the member joined the football club</td><td>StartDate</td><td>DATE</td></tr><tr><td>1345</td><td>The number of points a member has earned. Members of the club earn points for different activities.</td><td>Points</td><td>INTEGER</td></tr></table> | Example value | Explanation | Variable name | Data type | "Wong" | The preferred name of the member joining the football club | MemberName | STRING | FALSE | A value to indicate whether an existing member of the club lives at the same address | FamilyMember | BOOLEAN | 19/02/1983 | When the member joined the football club | StartDate | DATE | 1345 | The number of points a member has earned. Members of the club earn points for different activities. | Points | INTEGER | 4 |
|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------|---------------|-----------|--------|------------------------------------------------------------|------------|--------|-------|--------------------------------------------------------------------------------------|--------------|---------|------------|------------------------------------------|-----------|------|------|-----------------------------------------------------------------------------------------------------|--------|---------|---|
| Example value                                              | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Variable name | Data type   |               |           |        |                                                            |            |        |       |                                                                                      |              |         |            |                                          |           |      |      |                                                                                                     |        |         |   |
| "Wong"                                                     | The preferred name of the member joining the football club                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | MemberName    | STRING      |               |           |        |                                                            |            |        |       |                                                                                      |              |         |            |                                          |           |      |      |                                                                                                     |        |         |   |
| FALSE                                                      | A value to indicate whether an existing member of the club lives at the same address                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | FamilyMember  | BOOLEAN     |               |           |        |                                                            |            |        |       |                                                                                      |              |         |            |                                          |           |      |      |                                                                                                     |        |         |   |
| 19/02/1983                                                 | When the member joined the football club                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | StartDate     | DATE        |               |           |        |                                                            |            |        |       |                                                                                      |              |         |            |                                          |           |      |      |                                                                                                     |        |         |   |
| 1345                                                       | The number of points a member has earned. Members of the club earn points for different activities.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Points        | INTEGER     |               |           |        |                                                            |            |        |       |                                                                                      |              |         |            |                                          |           |      |      |                                                                                                     |        |         |   |
| One mark for each appropriate variable name plus data type |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |             |               |           |        |                                                            |            |        |       |                                                                                      |              |         |            |                                          |           |      |      |                                                                                                     |        |         |   |

## 72. M/J 21/P23/Q2/c

2(c)

Example solution:

6



One mark for each functional group as listed below:

|      |                                                                                                                                                                                                                                                                                                                                                                                                             |  |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 2(c) | <p>Explanation of mark points:</p> <ol style="list-style-type: none"> <li>1 Initialise MAX to first value input</li> <li>2 Set Total to zero</li> <li>3 Input 49 more values (or 50 values in total)</li> <li>4 Sum all values input</li> <li>5 Set new MAX when Input value &gt; MAX <b>in a loop</b></li> <li>6 Sum all but largest (or subtract MAX from total), calculate and output average</li> </ol> |  |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

**73. M/J 21/P23/Q7/b**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 7(b) | <p>Marks awarded for any reference to each of the following steps of the algorithm:</p> <ol style="list-style-type: none"> <li>1 Mention of variable for use as array index</li> <li>2 Use of a loop (to check through the array)</li> <li>3 If word is the same as the current array element then return FALSE / set flag</li> <li>4 If word not already in array, loop to find unused element (<i>second loop</i>)</li> <li>5 Store word in unused element and return TRUE, otherwise return FALSE</li> </ol> <p>VARIATION:</p> <ol style="list-style-type: none"> <li>1 Mention of variable for use as array index</li> <li>2 Use of a loop (to check through the array)</li> <li>3 Save index of (first) unused element found</li> <li>4 If word is the same as the current array element then return FALSE / set flag</li> <li>5 If word not already in array <b>and</b> unused element available, store word in unused element and return TRUE otherwise return FALSE</li> </ol> <p><b>Note: Max 4 marks</b></p> | <b>4</b> |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|